

CS-663

Digital Image Processing (DIP)

Detecting Edges and Corners

Suyash P. Awate

Edge

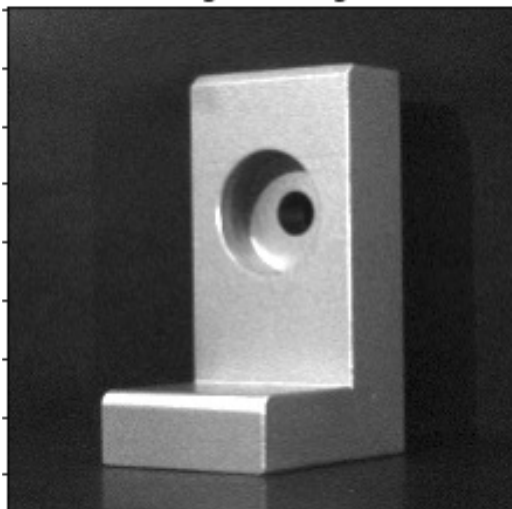
- Edge: locations of sharp change in image intensity/color over space
- Cause: discontinuities in depth, discontinuities in surface orientation, change in material properties, variation in scene illumination, etc.
- Edge detection can lead to abstract representation of image



Edge Detection: Sobel

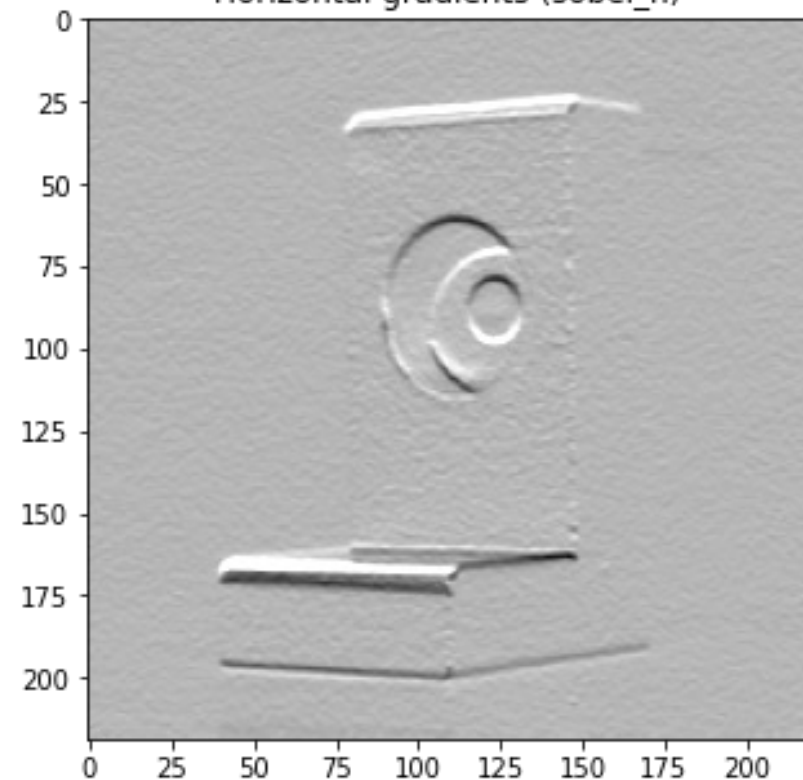
- Sobel operator (we saw before) gives smoothed gradients

Original image

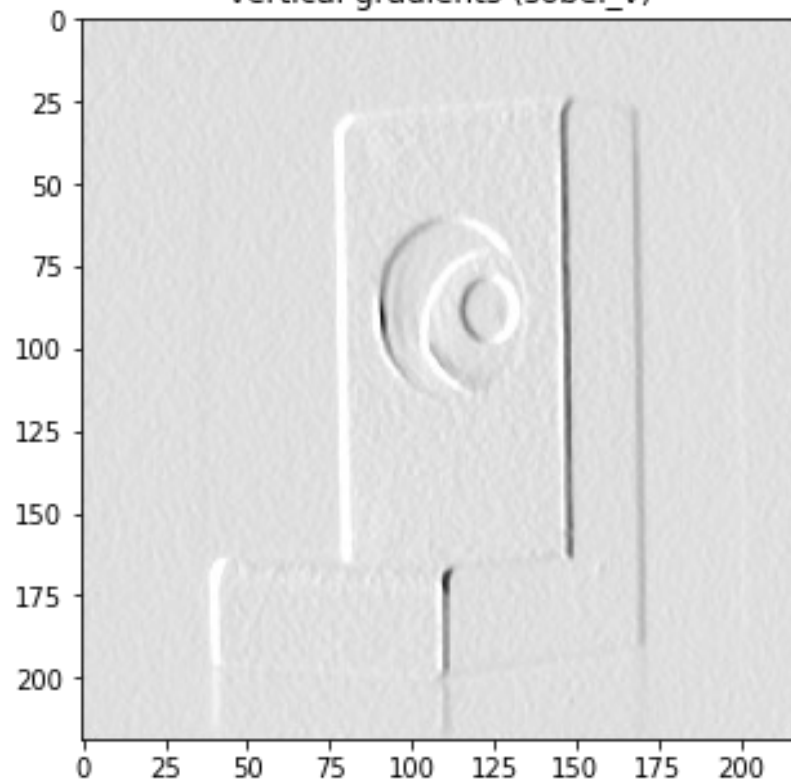


$$\mathbf{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} * \mathbf{A}$$
$$\mathbf{G}_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} * \mathbf{A}$$

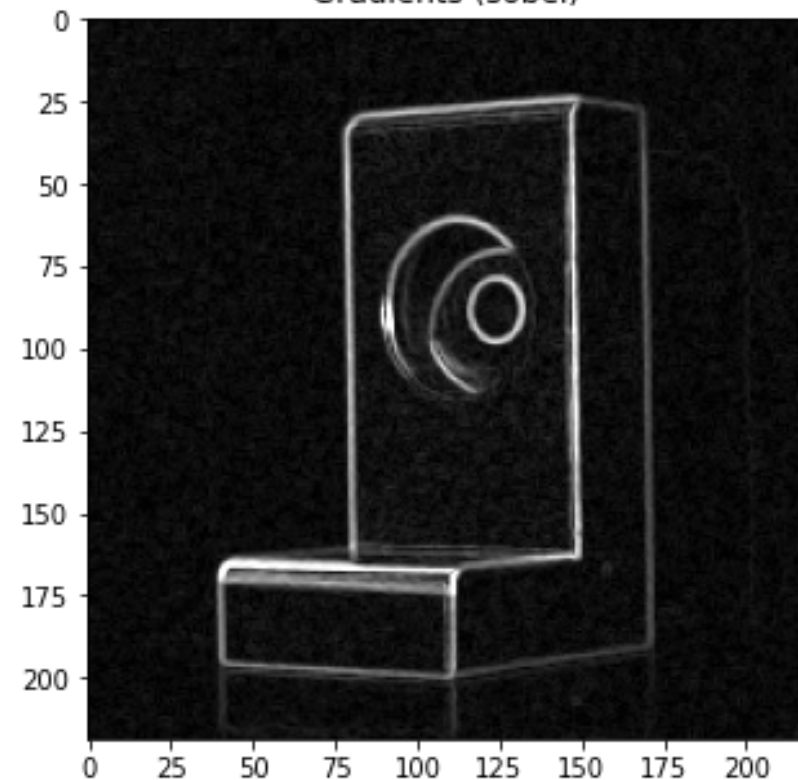
Horizontal gradients (sobel_h)



vertical gradients (sobel_v)



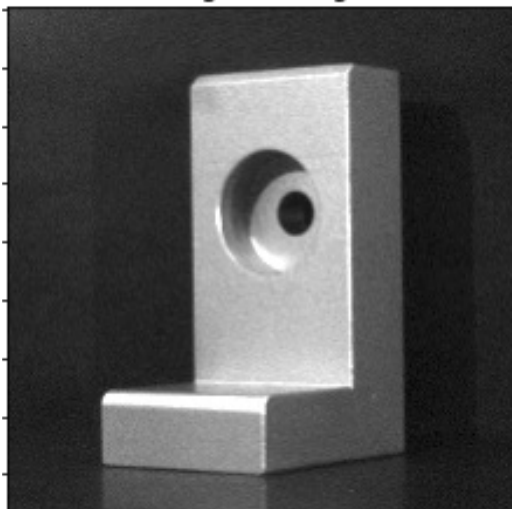
Gradients (sobel)



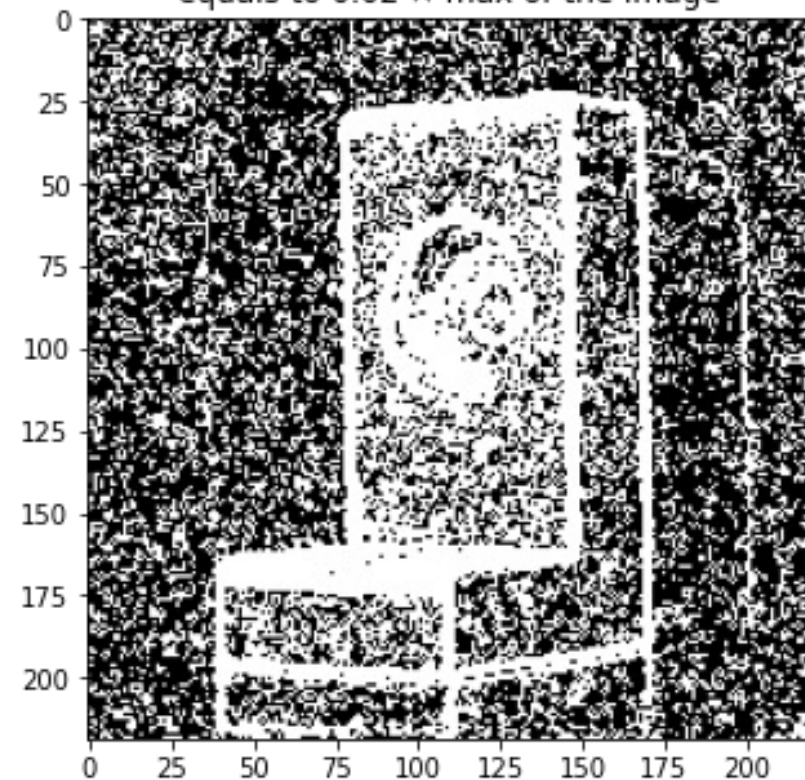
Edge Detection: Sobel

- Sobel **detector** introduces **thresholds** on gradients to detect edges

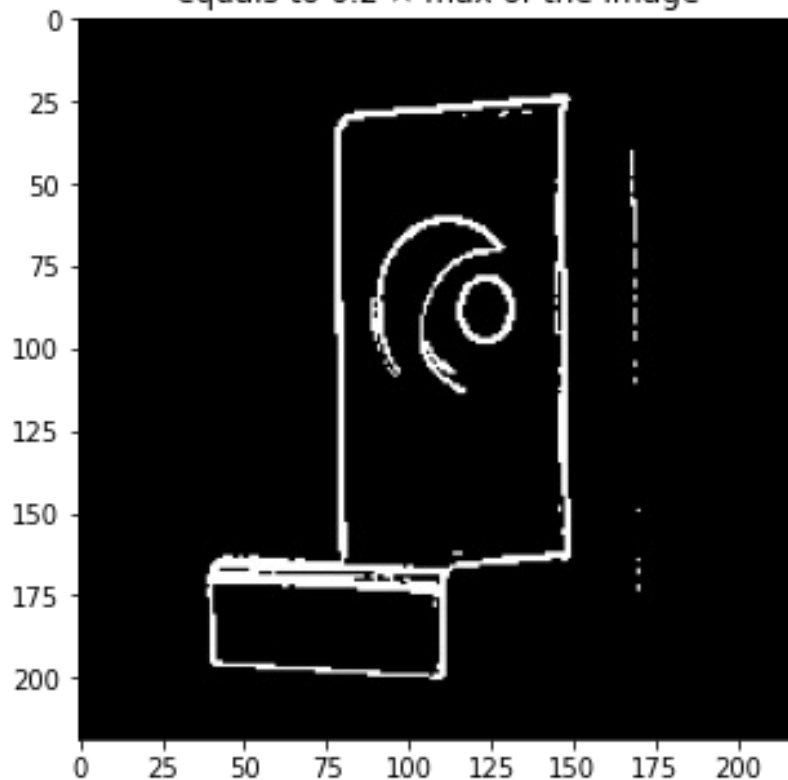
Original image



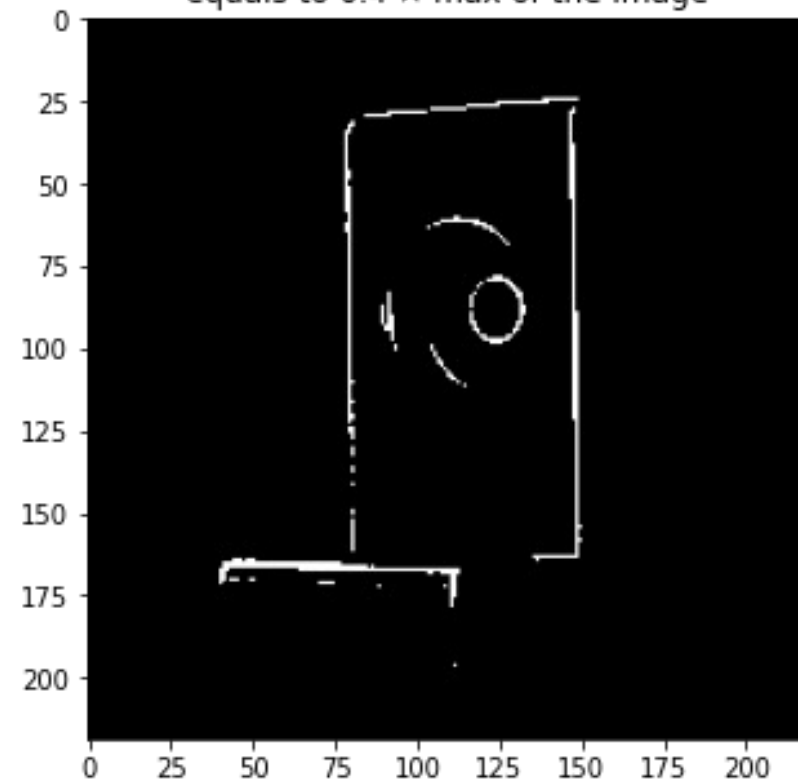
Sobel detector with threshold equals to $0.02 \times \text{max of the image}$



Sobel detector with threshold equals to $0.2 \times \text{max of the image}$



Sobel detector with threshold equals to $0.4 \times \text{max of the image}$



$$\mathbf{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} * \mathbf{A}$$
$$\mathbf{G}_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} * \mathbf{A}$$

Edge Detection: Marr-Hildreth

Proc. R. Soc. Lond. B 207, 187–217 (1980)

Printed in Great Britain

Theory of edge detection

BY D. MARR AND E. HILDRETH

*M.I.T. Psychology Department and Artificial Intelligence Laboratory,
79 Amherst Street, Cambridge, Massachusetts 02139, U.S.A.*

(Communicated by S. Brenner, F.R.S. – Received 22 February 1979)

Edge Detection: Marr-Hildreth

- Algorithm

- 1) Convolve image with **LoG**

- Can be approximated as DoG

- 2) Detect **zero crossings**

- a) Compare signs of pixel intensities of neighbors

- Left versus right
- Up versus down
- NW versus SE
- NE versus SW

- b) If any comparison leads to **different signs** and **magnitude of difference > threshold**, then, edge found

Edge



1st derivative

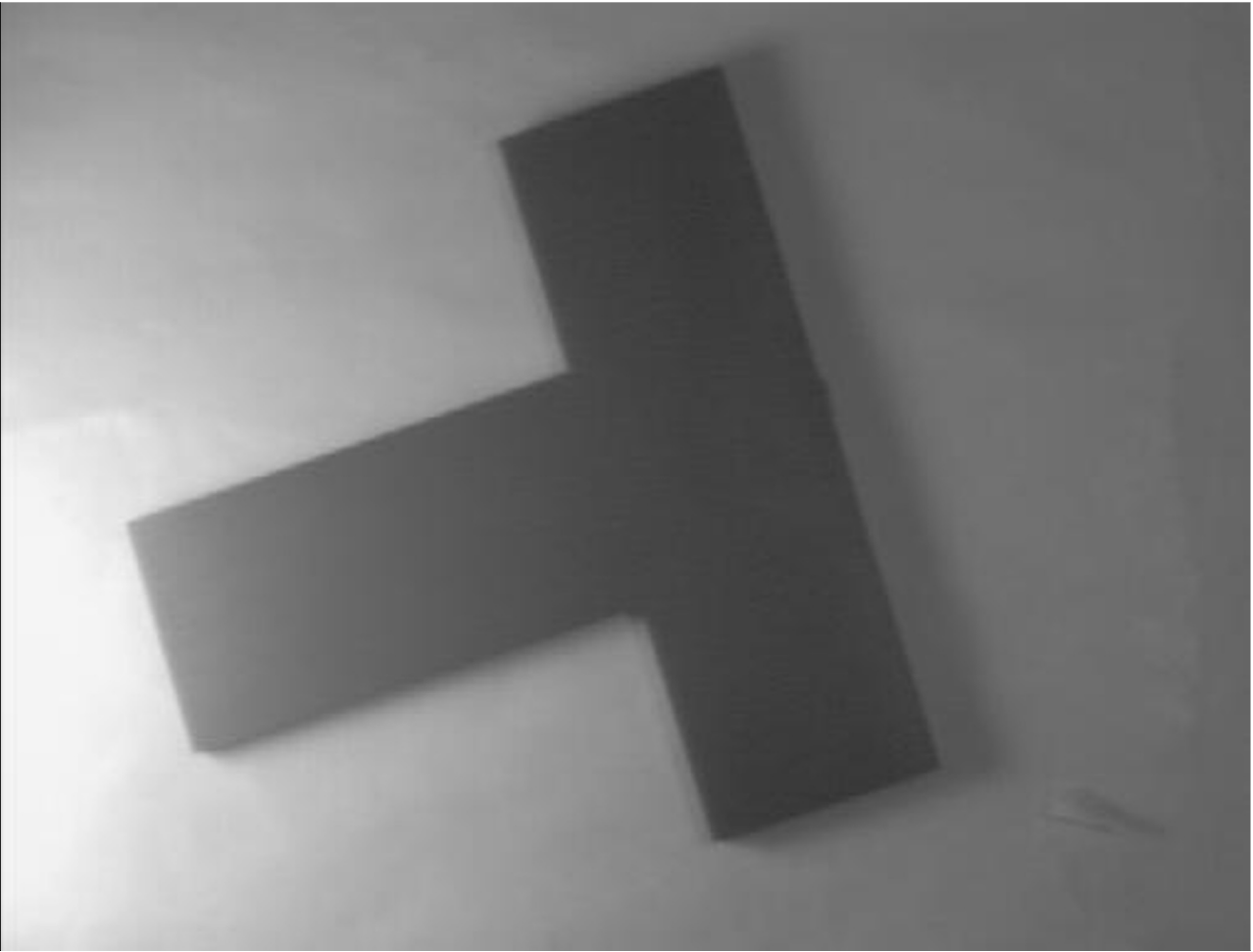
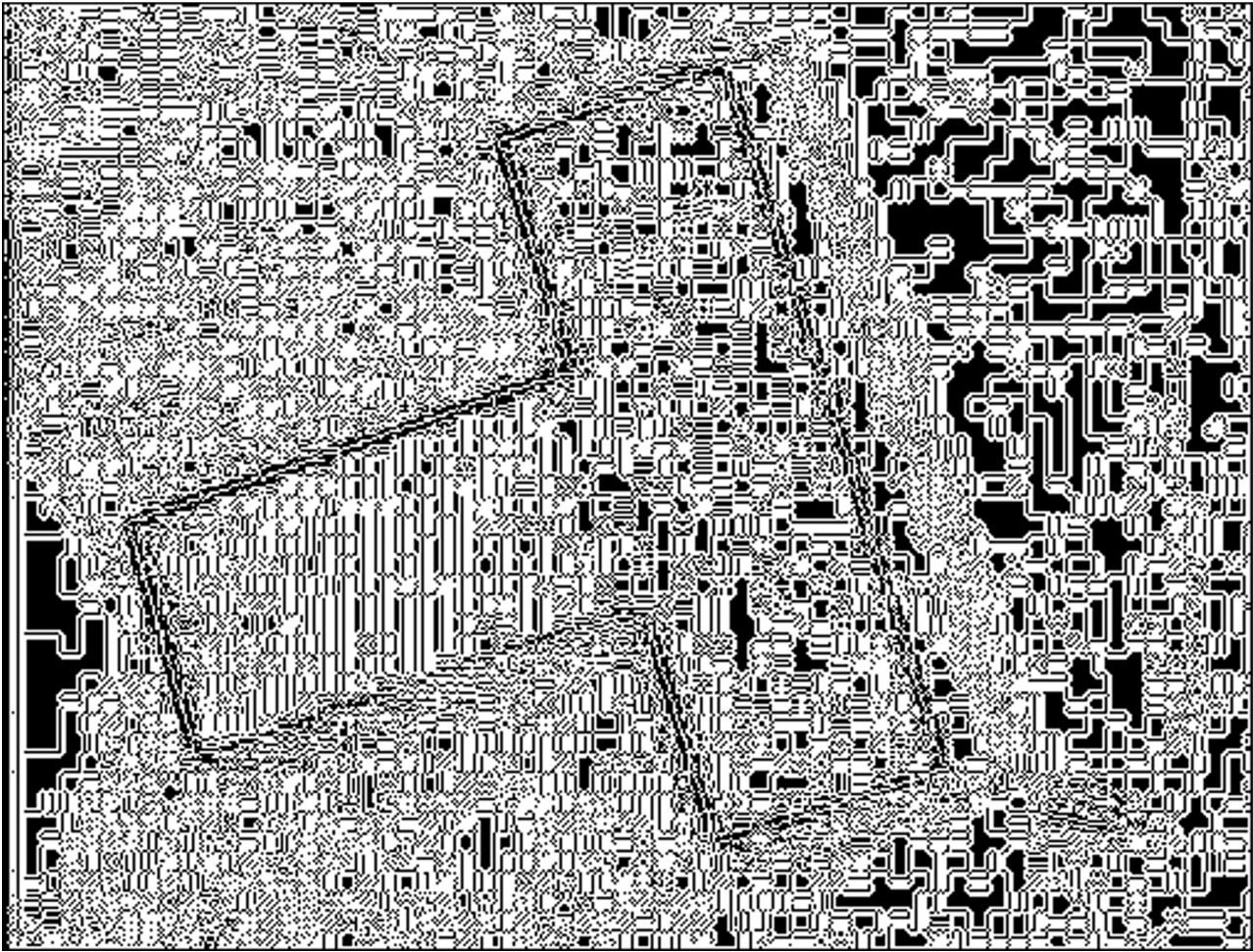


2nd derivative



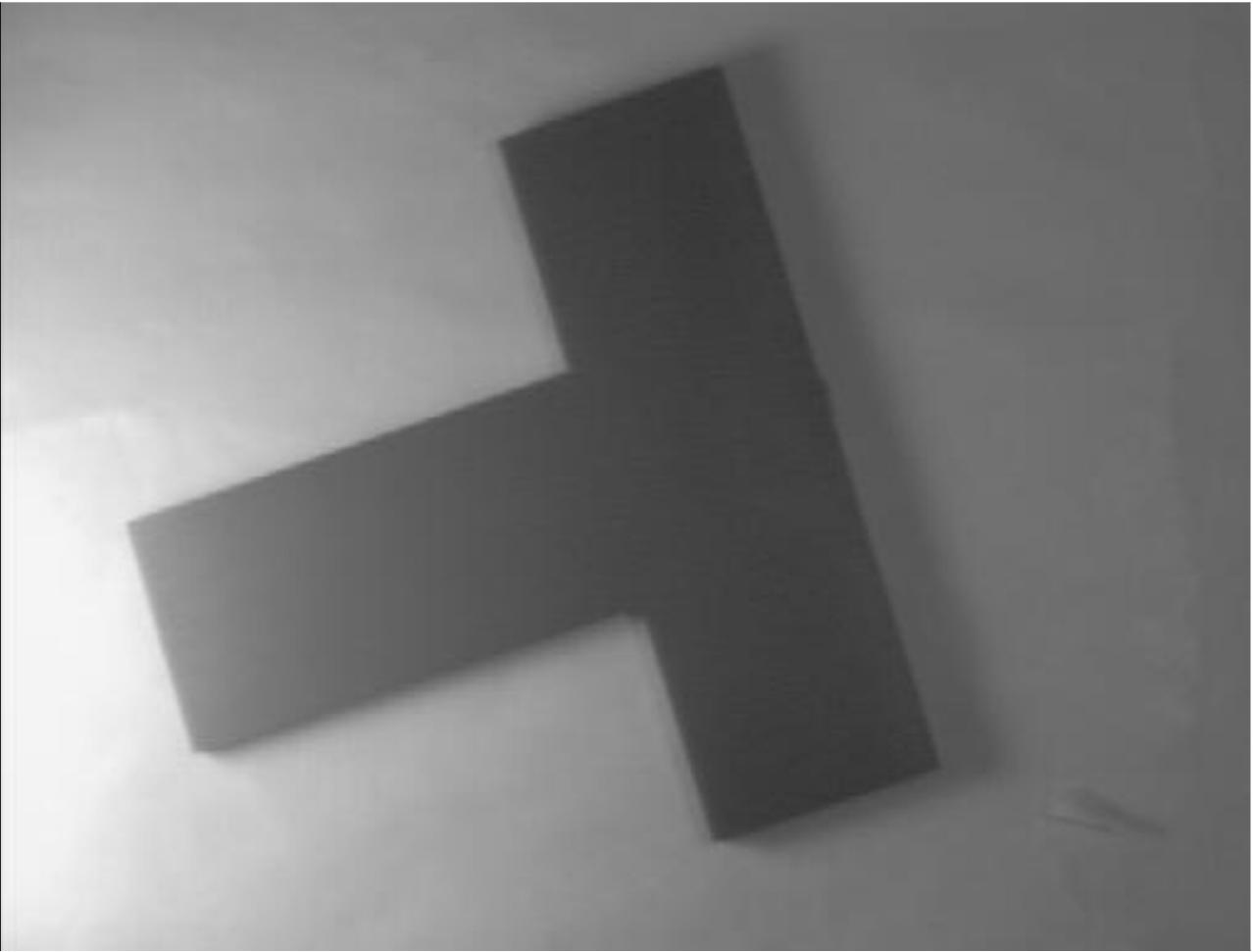
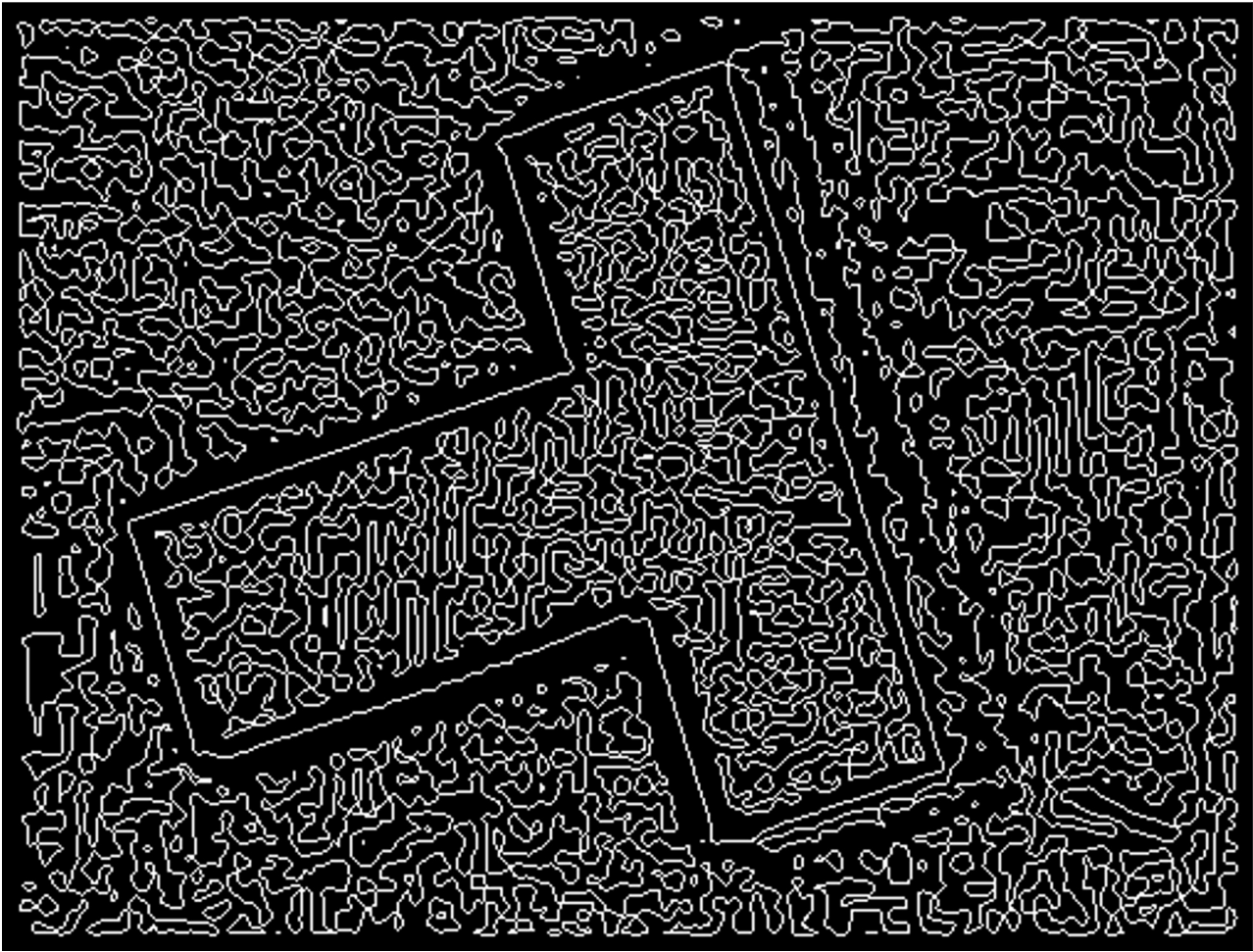
Edge Detection: Marr-Hildreth

- Gaussian variance (within LoG) $\rightarrow 0$
- Threshold = 0



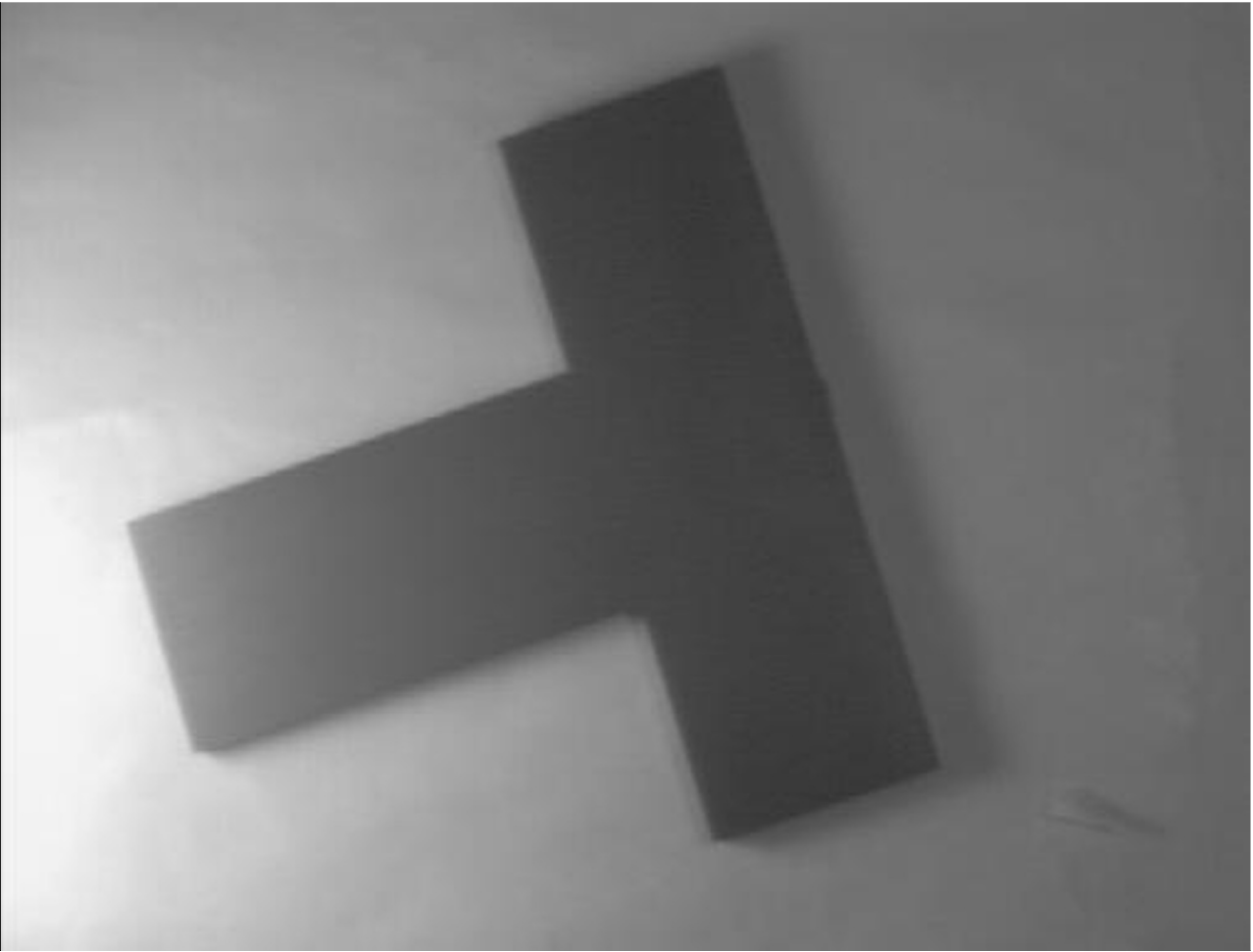
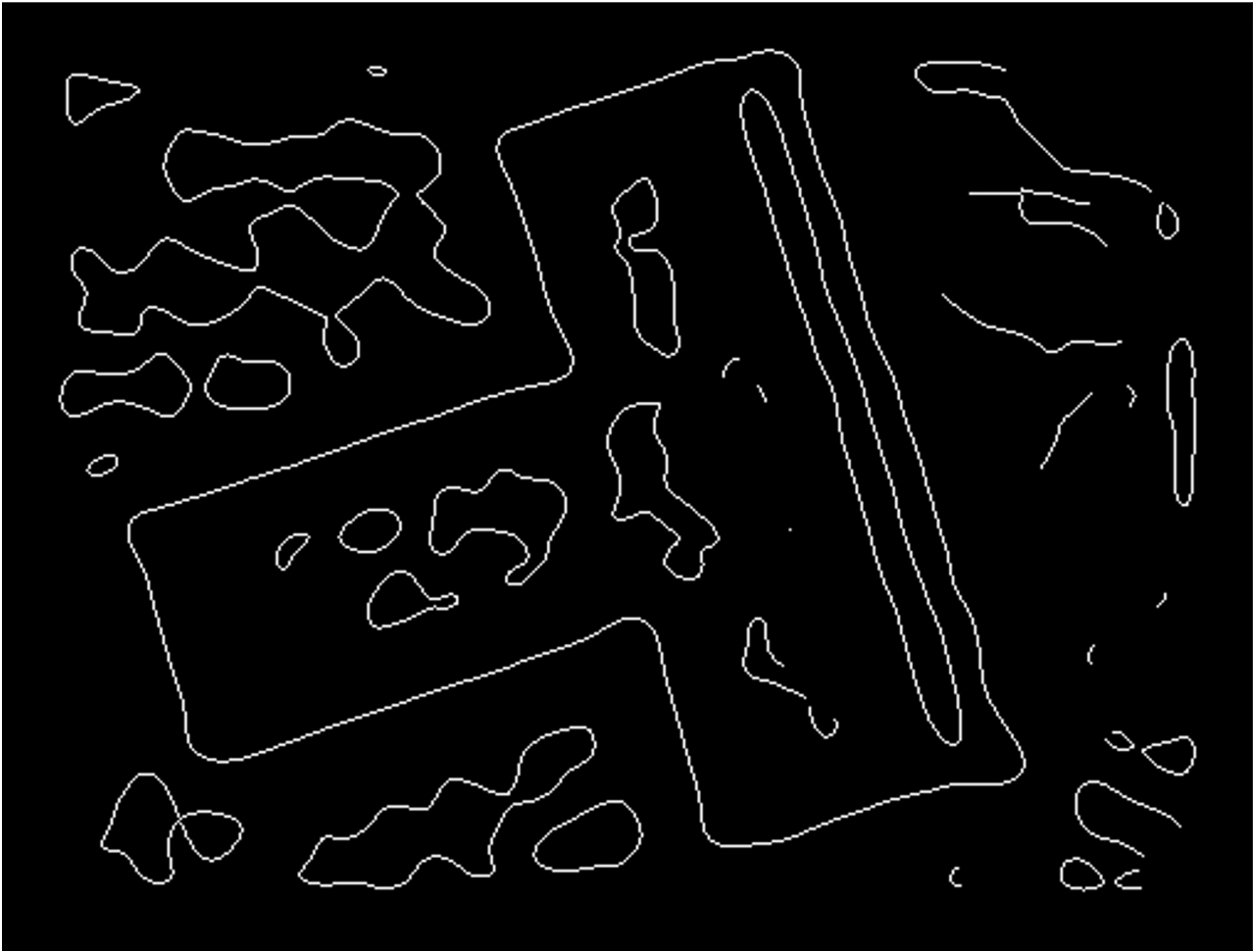
Edge Detection: Marr-Hildreth

- Gaussian variance (within LoG) $\sigma^2 > 0$
- Threshold = 0



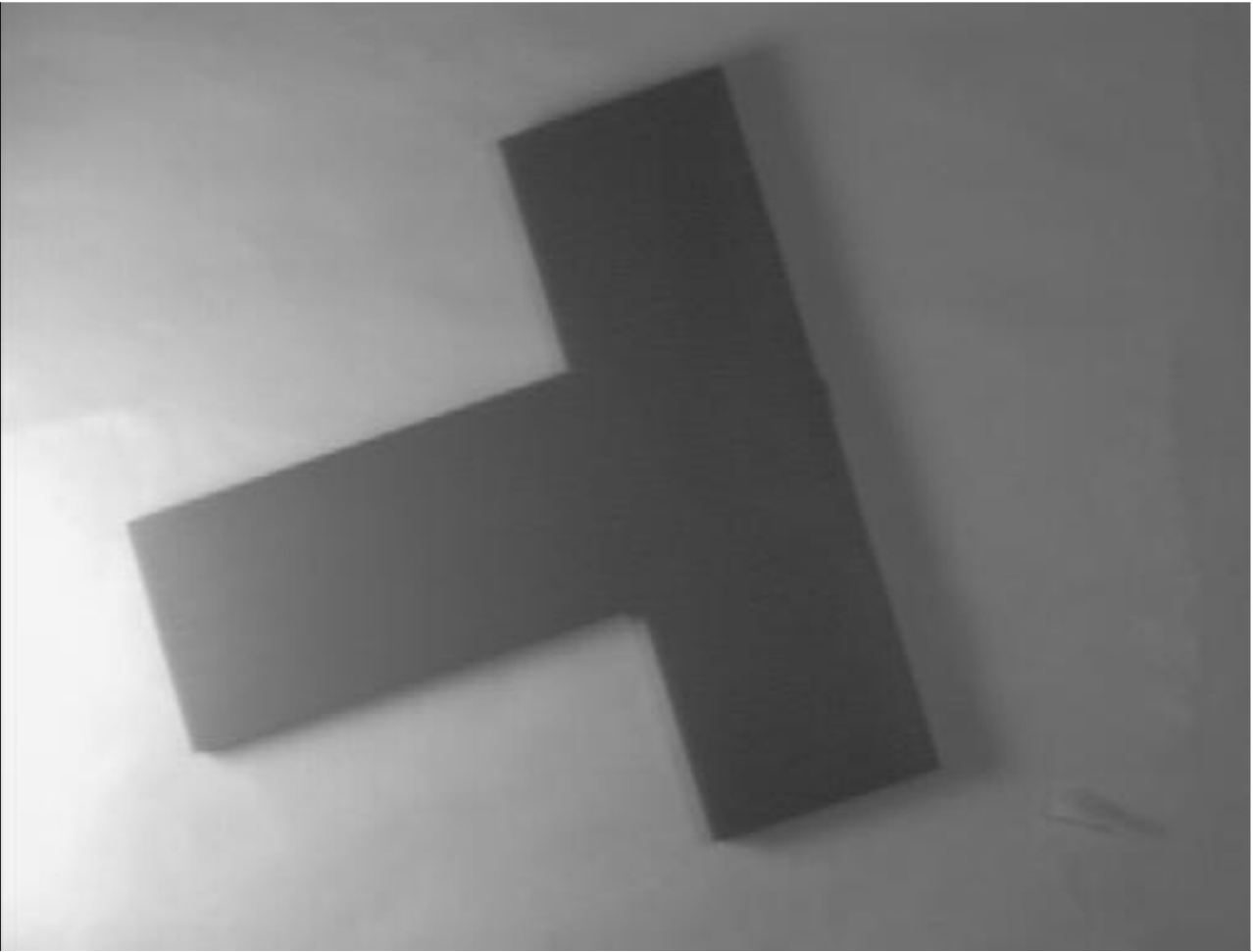
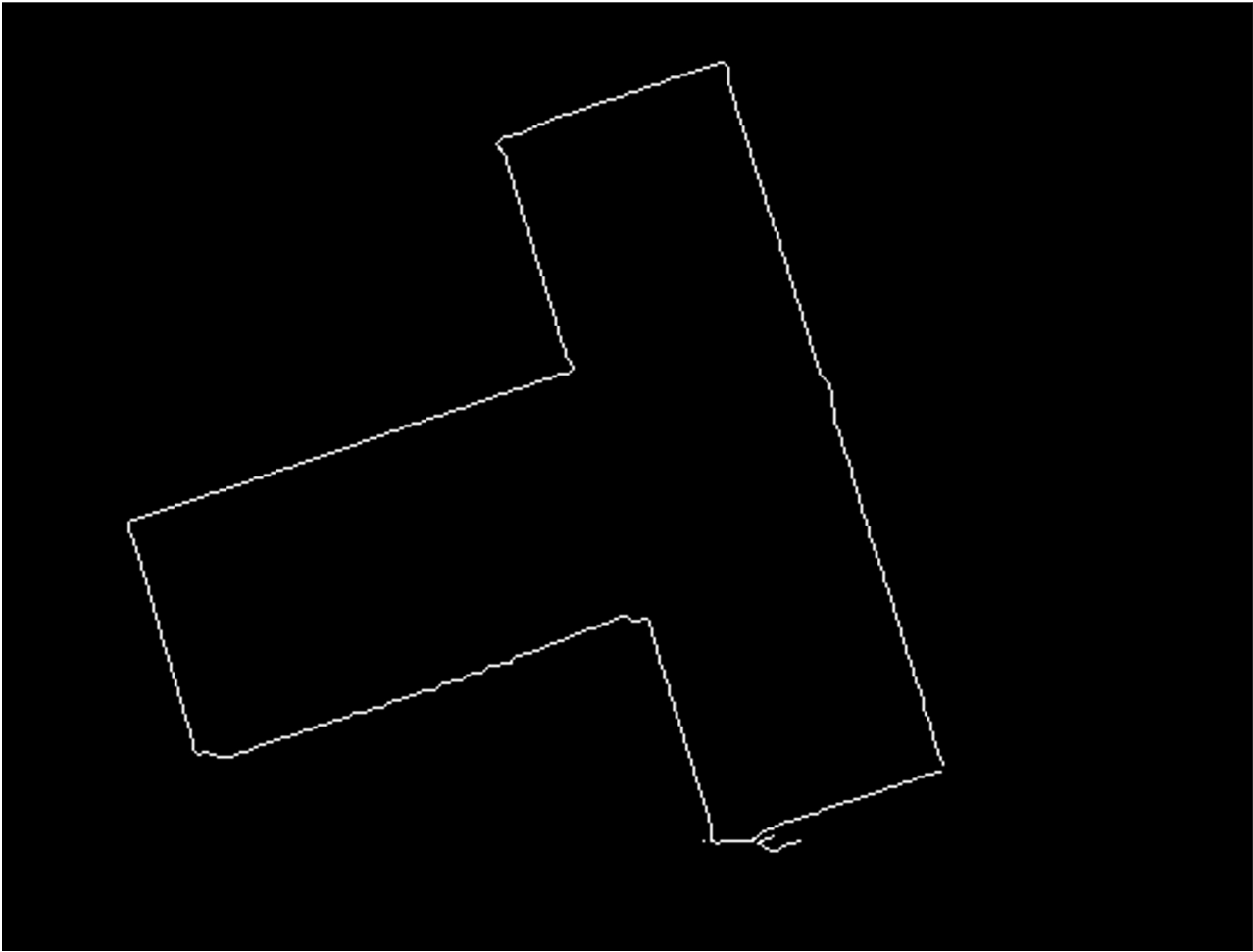
Edge Detection: Marr-Hildreth

- Gaussian variance (within LoG) $g_2 > g_1 > 0$
- Threshold = 0



Edge Detection: Marr-Hildreth

- Gaussian variance (within LoG) $g_3 > g_2 > 0$
- Threshold = 0



Edge Detection: Marr-Hildreth

- Left: original image
- Middle: LoG
- Right: edges
- [vincmazet.github.io/bip/detection/edges.html]



David Marr

- Neuroscientist, psychologist
- Studied computational models of learning in 3 neural networks in human brain: cerebellum (1969), hippocampus (1970), neocortex (1971)
- After PhD in physiology, he got interested in visual processing, AI
- Marr Prize: one of the most prestigious awards in computer vision
- Died of cancer at age 35

A general theory for cerebral cortex.

David Courtenay. Marr ; University of Cambridge. Department of Physiology.
University of Cambridge : Ph.D. Dissertation.



Edge Detection: Canny

- John Canny. 1986. Doesn't need 2nd derivatives.

- Algorithm

- 1) Gaussian smoothing to reduce noise

- 2) Gradient-magnitude G computation at each pixel

- +vector and -vector are equivalent

$$G = \sqrt{G_x^2 + G_y^2}$$
$$\Theta = \text{atan2}(G_y, G_x)$$

- 3) Edge **thinning** by **non-maximum suppression**: see next ...

- 4) Edge **tracing** via **hysteresis thresholding**: see next ...

A Computational Approach to Edge Detection

Author:  [J_Canny](#) | [Authors Info & Claims](#)

Edge Detection: Canny

- Algorithm

- 1) Gaussian smoothing to reduce noise

- 2) At each pixel location “p”, compute gradient vector $v(p)$

- 3) Edge thinning by non-maximum suppression**

- At location “p”, let unit vector along gradient be $u(p) := v(p) / \|v(p)\|_2$

- Right: image of gradient magnitudes $\|v(p)\|_2$

- For pixel locations “p” where

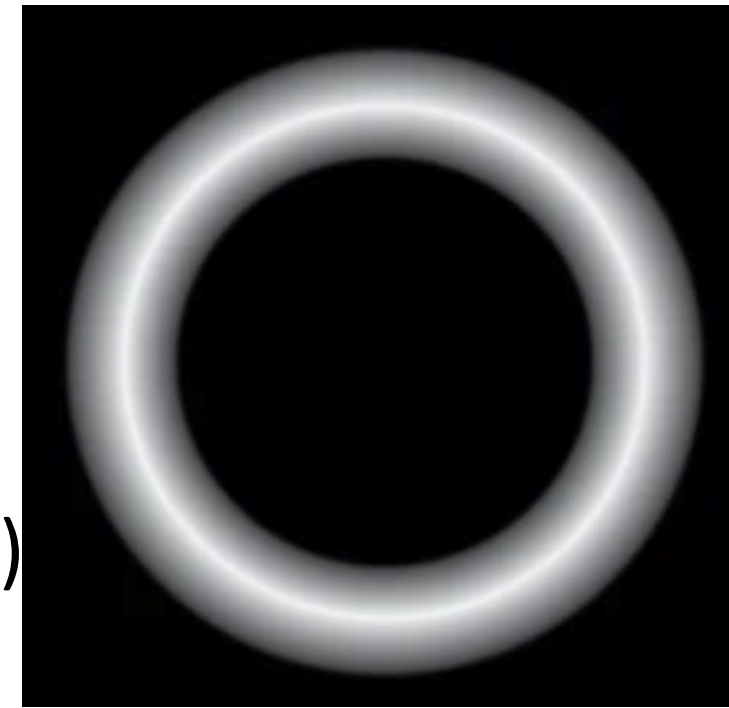
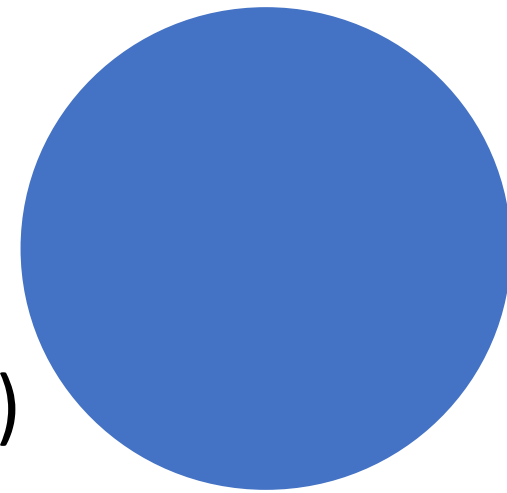
 - $\|v(p)\|_2 > \|v(p+u(p))\|_2$ and

 - $\|v(p)\|_2 > \|v(p-u(p))\|_2$,

 - select pixel as candidate pixel on edge;**

 - else suppress pixel (in image on right: set value = 0)

 - Needs interpolation at $p \pm u(p)$



Edge Detection: Canny

- Algorithm

- 1) Gaussian smoothing to reduce noise

- 2) At each pixel location “p”, compute gradient vector $v(p)$

- 3) Edge thinning by non-maximum suppression

- 4) Edge tracing via hysteresis thresholding**

- Single threshold on $\|v(p)\|_2$ isn't effective in reliably detecting pixels on edges
- Define 2 thresholds: **thresholdHigh**>0 and **thresholdLow**>0
- First, select **candidate pixels where $\|v(p)\|_2 > \text{thresholdHigh}$** as surely belonging to an edge
- Then, find all connected paths starting from pixels selected above such that all **candidate pixels in path have $\|v(p)\|_2 > \text{thresholdLow}$** ; select all such pixel locations “p” on such paths as also belonging to an edge

Edge Detection: Canny

- Hysteresis thresholding makes following assumptions
- Consider an image where **value at pixel** location “p” is set to **magnitude of gradient** at “p”
 - 1) If **pixelValue > thresholdHigh**
→ pixel on edge
 - 2) If **pixelValue ≤ thresholdLow**
→ pixel NOT on edge
 - 3) If **thresholdLow < pixelValue ≤ thresholdHigh**
→ pixel on edge if
some connected path exists to pixel with pixelValue > thresholdHigh,
such that all pixels on path have values > thresholdLow

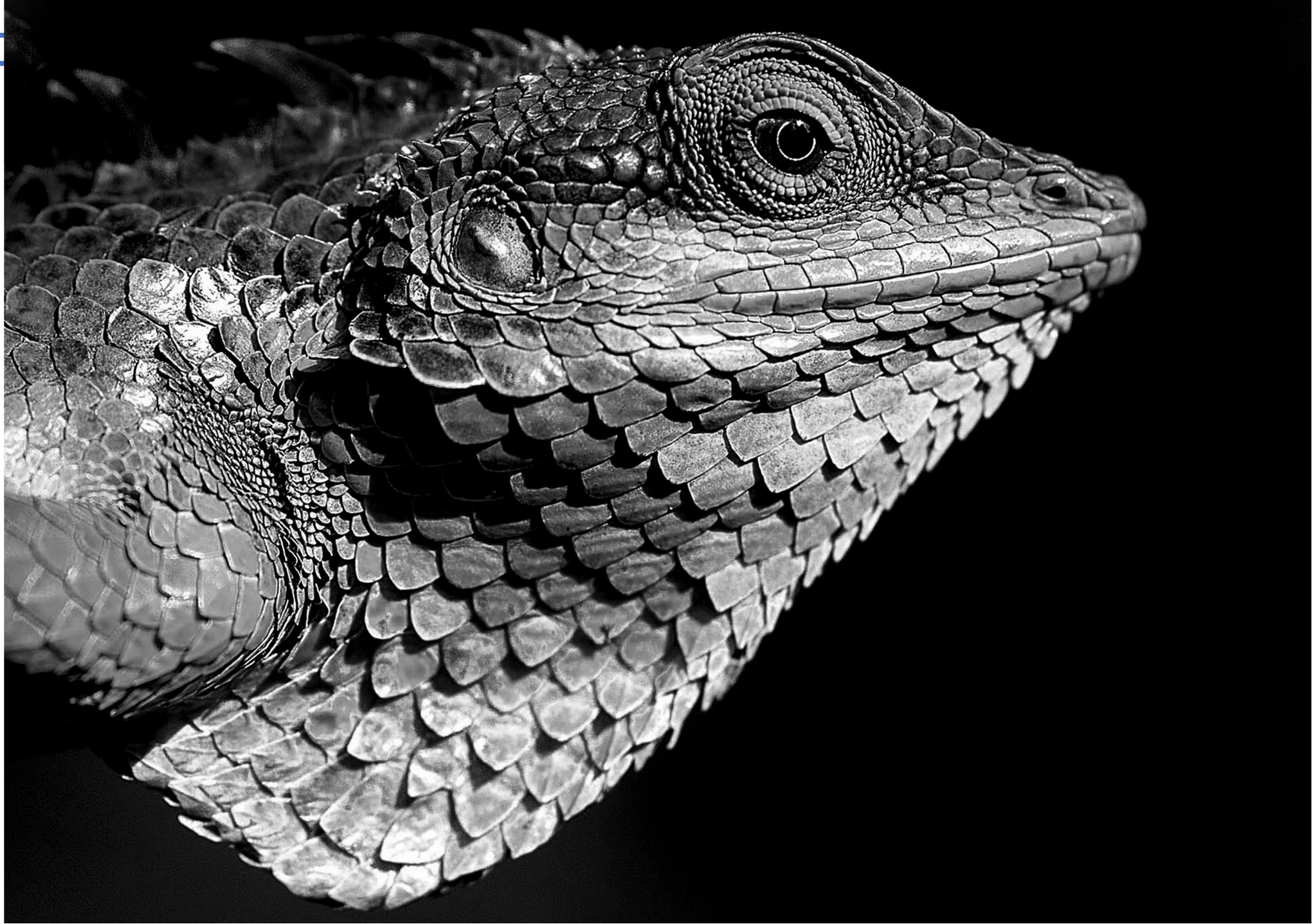
Edge Detect

- Canny edge detector
- Input image
 - Large Scaled Forest Lizard, from Valparai, Near Pollachi, Tamilnadu



Edge Detect

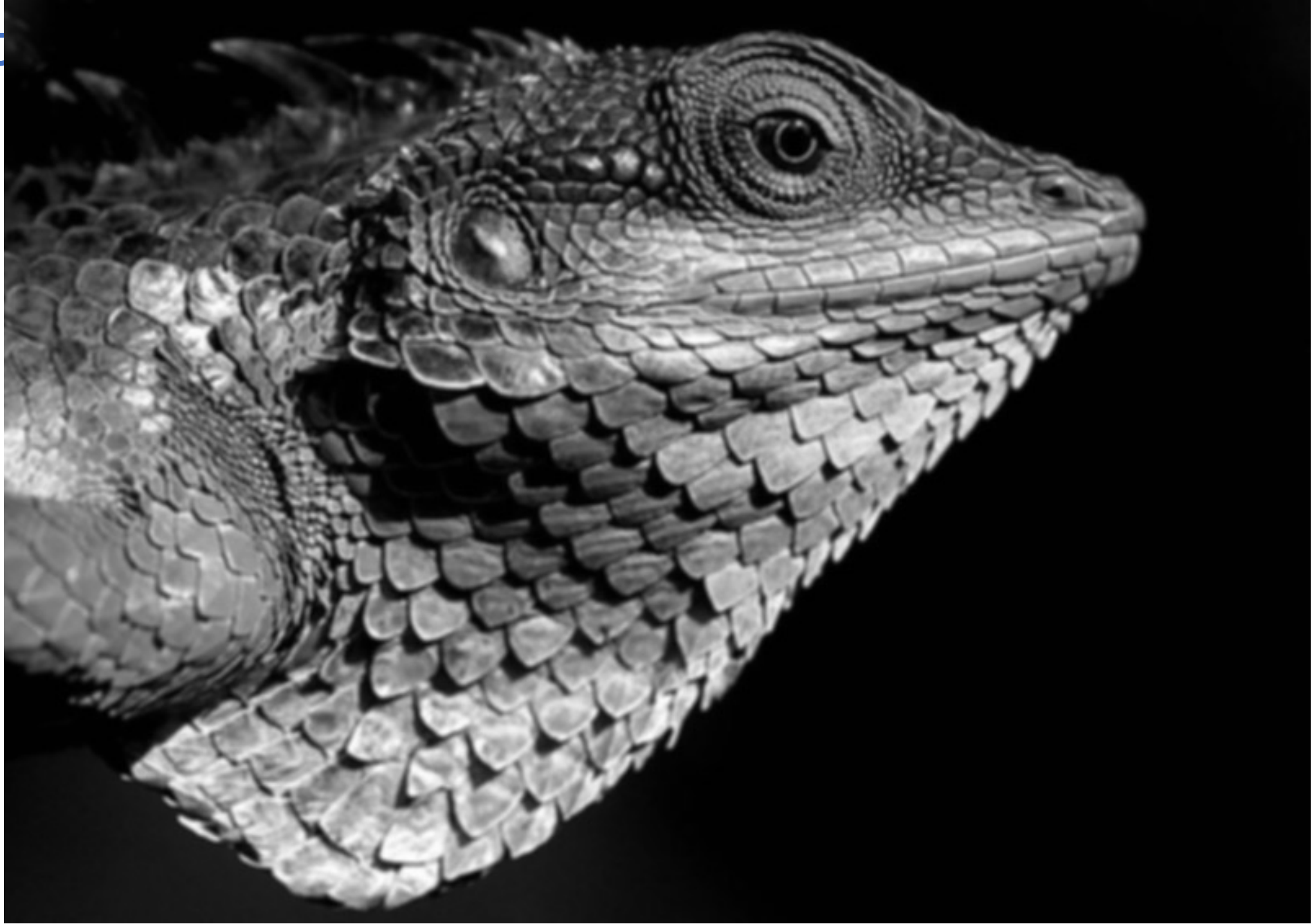
- Canny edge detector
- Input image
 - Large Scaled Forest Lizard, from Valparai, Near Pollachi, Tamilnadu



Edge Detect

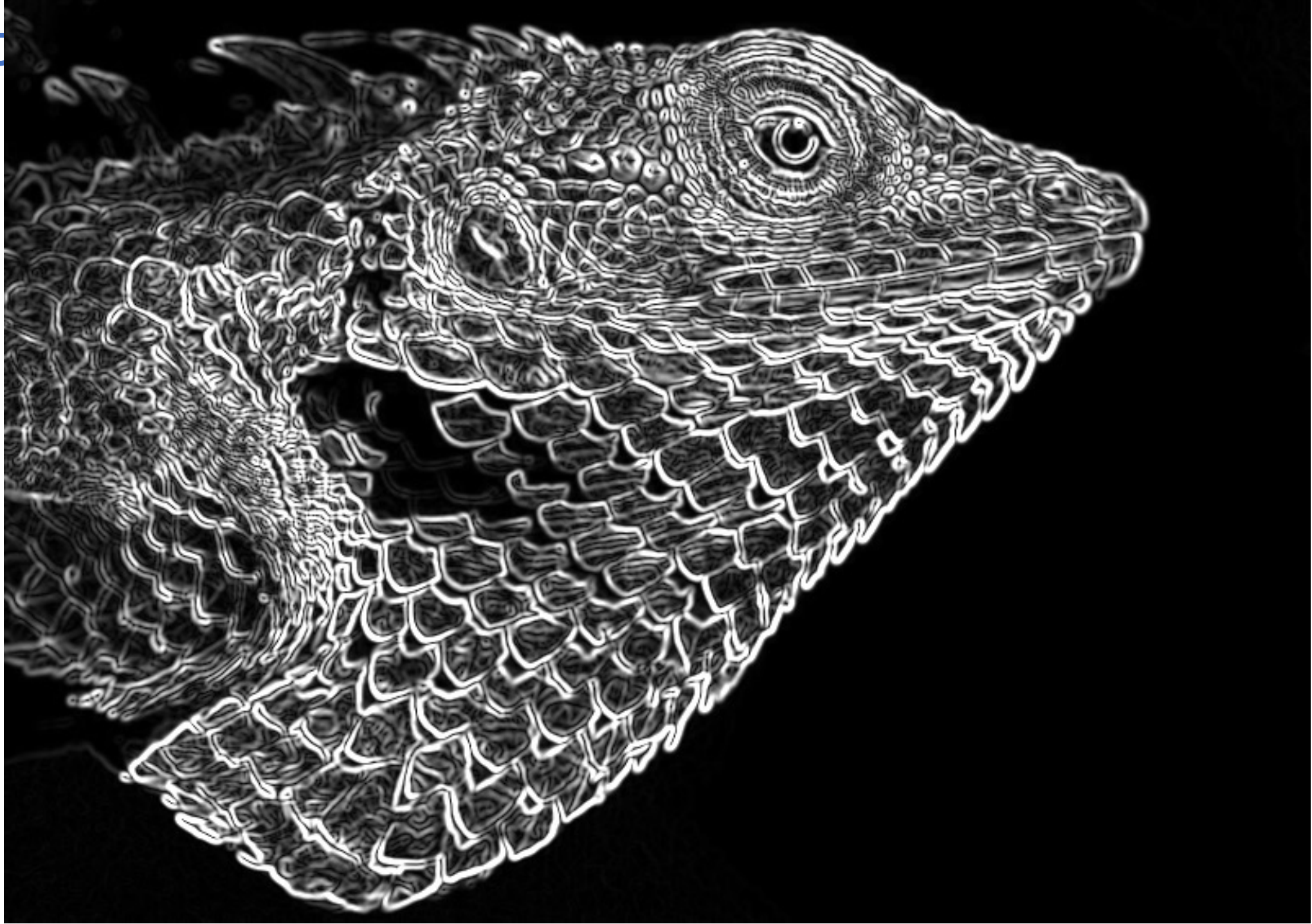
- Canny edge detector

1) Gaussian blur



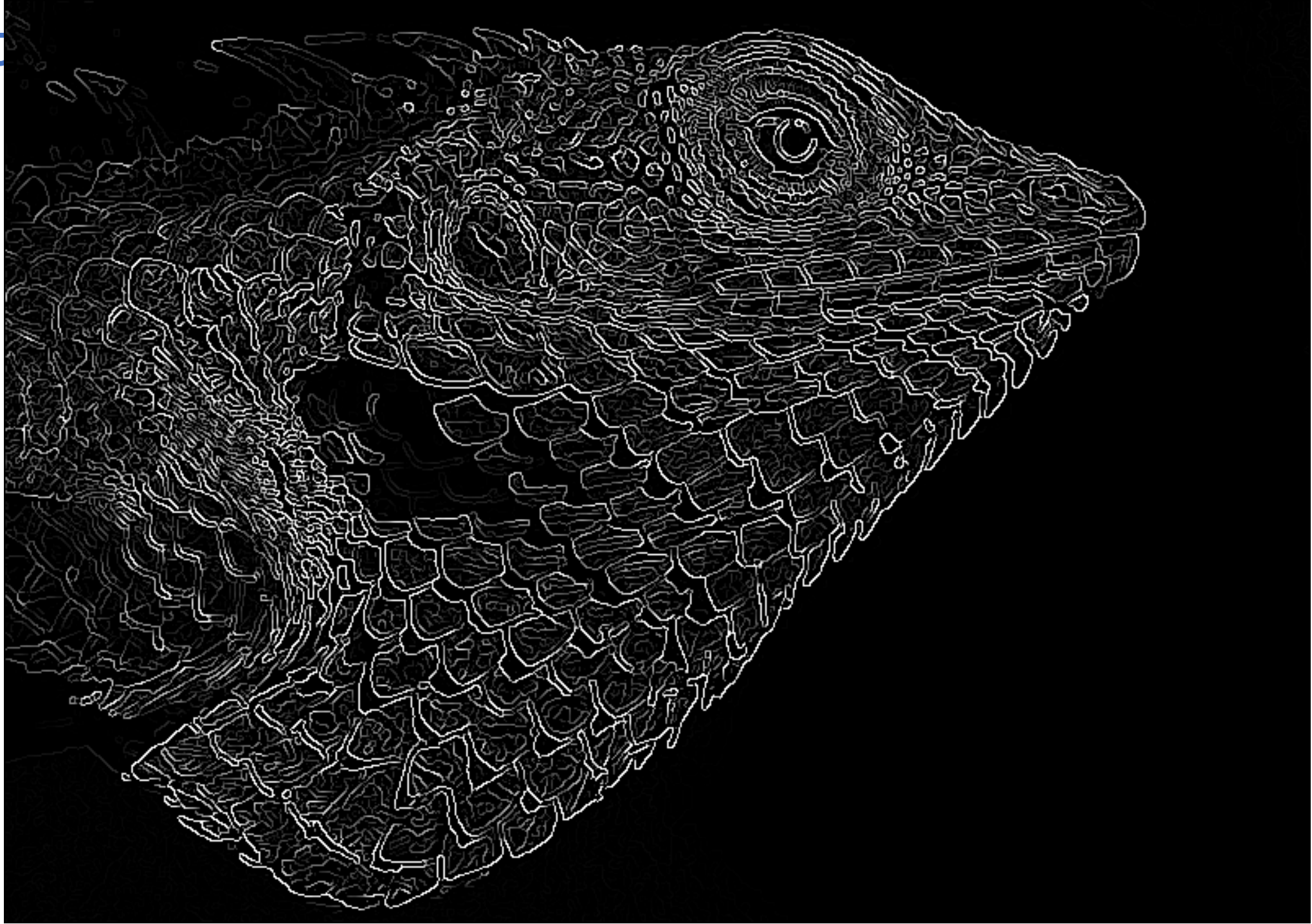
Edge Detect

- Canny edge detector
 - 1) Gaussian blur
 - 2) Intensity grad. mag.



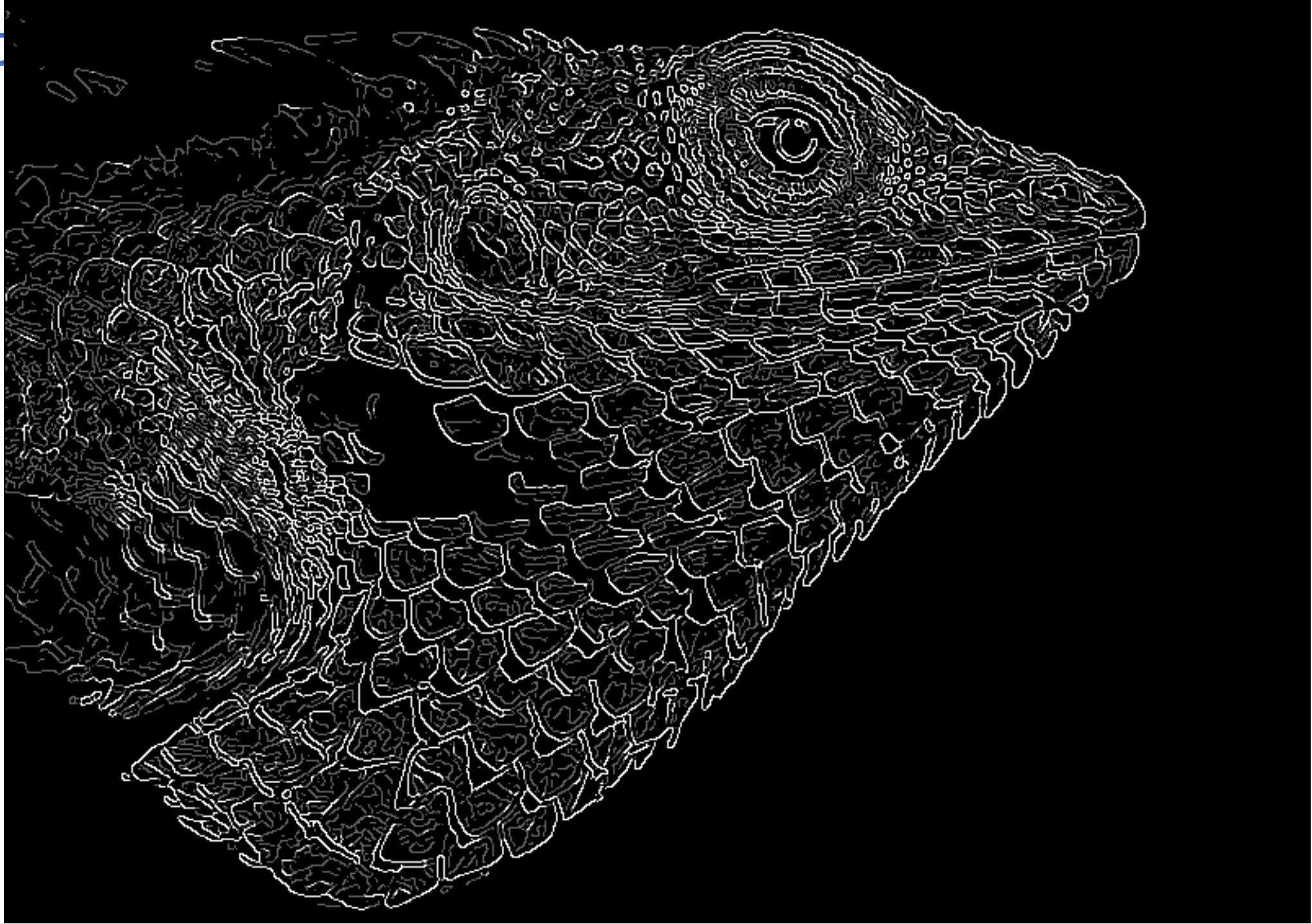
Edge Detect

- Canny edge detector
 - 1) Gaussian blur
 - 2) Intensity grad. mag.
 - 3) Non-max suppress



Edge Detect

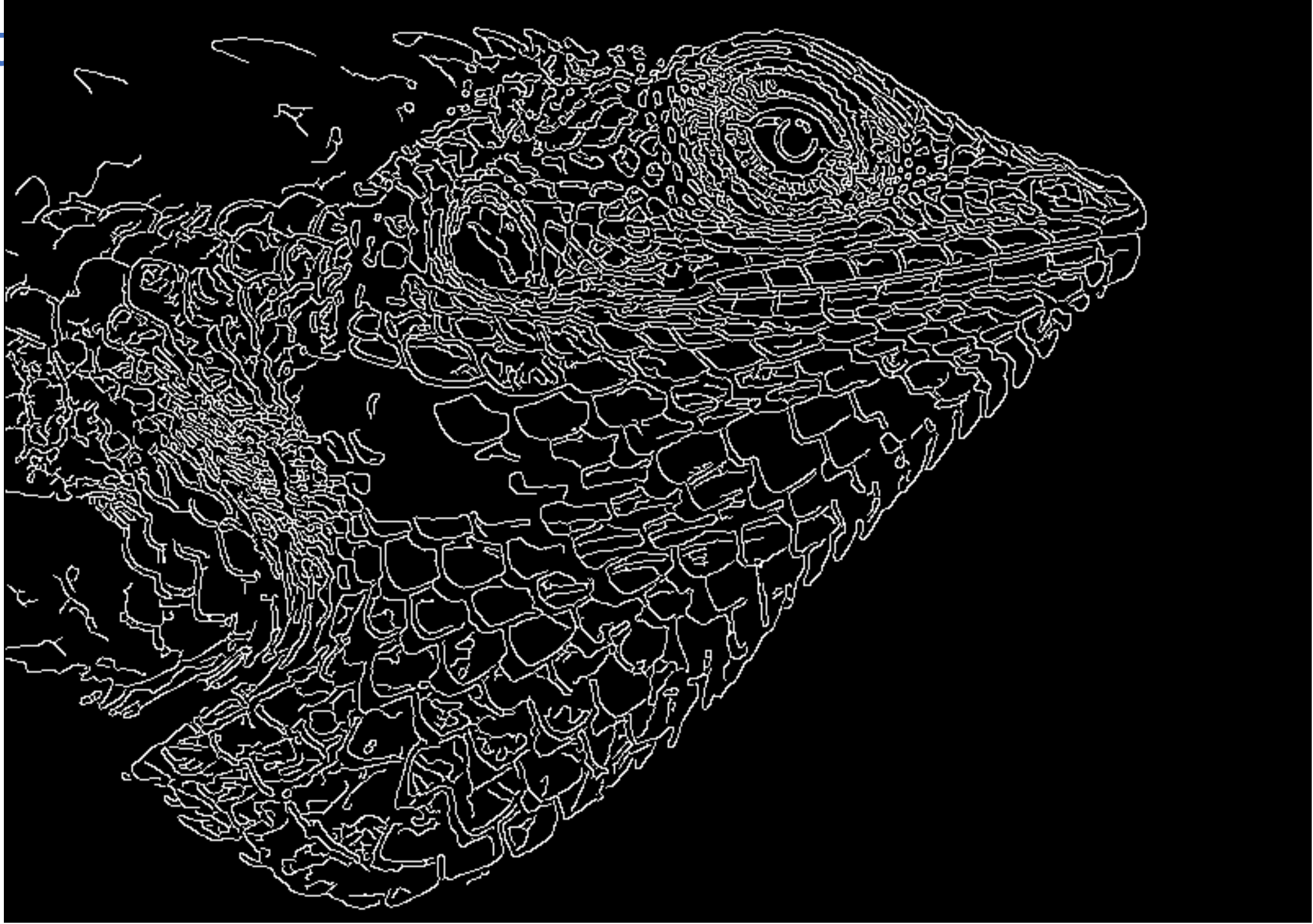
- Canny edge detector
 - 1) Gaussian blur
 - 2) Intensity grad. mag.
 - 3) Non-max suppress
 - 4) Double threshold



Edge Detect

- Canny edge detector

- 1) Gaussian blur
- 2) Intensity grad. mag.
- 3) Non-max suppress
- 4) Double threshold
- 5) Hysteresis

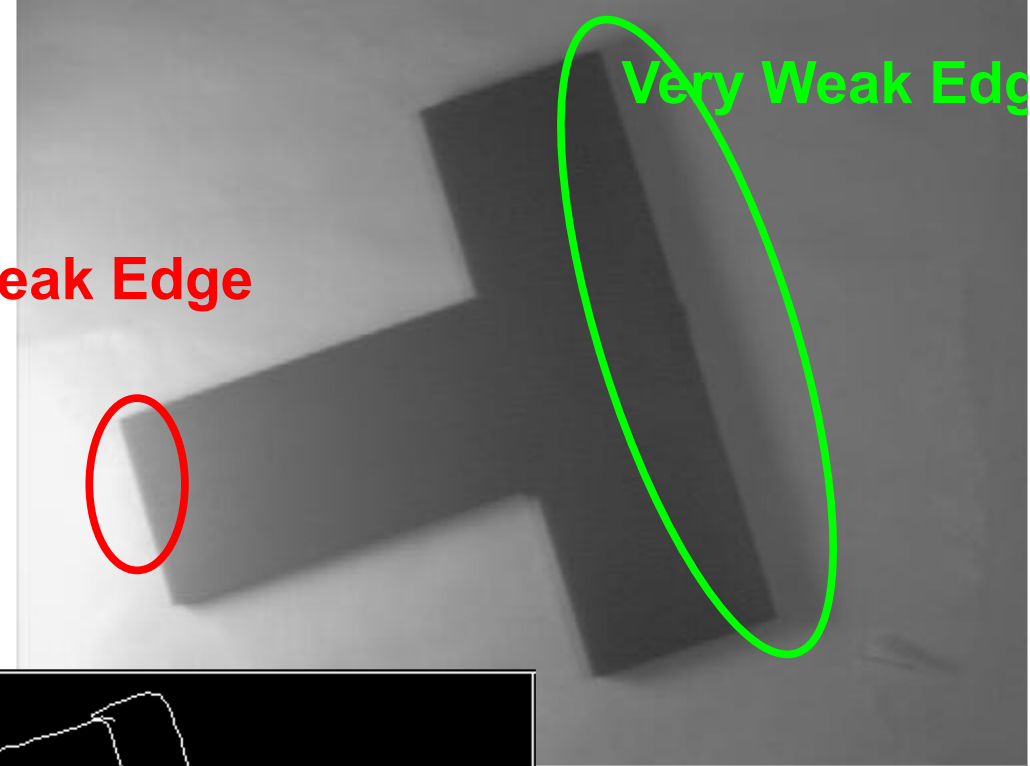
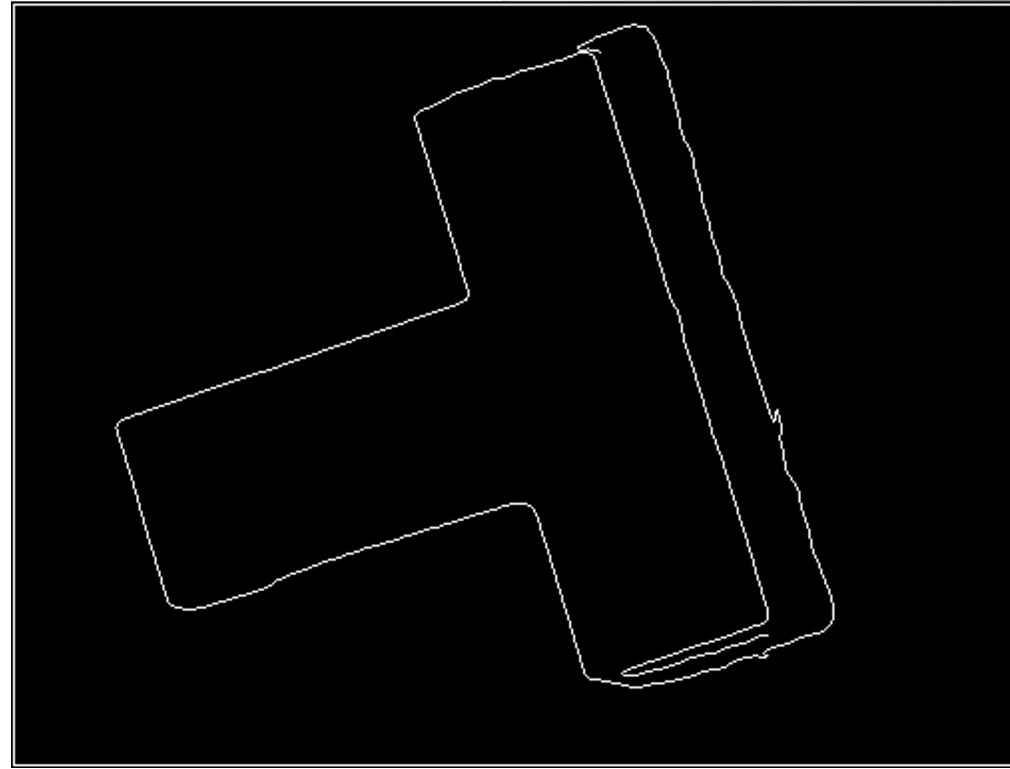
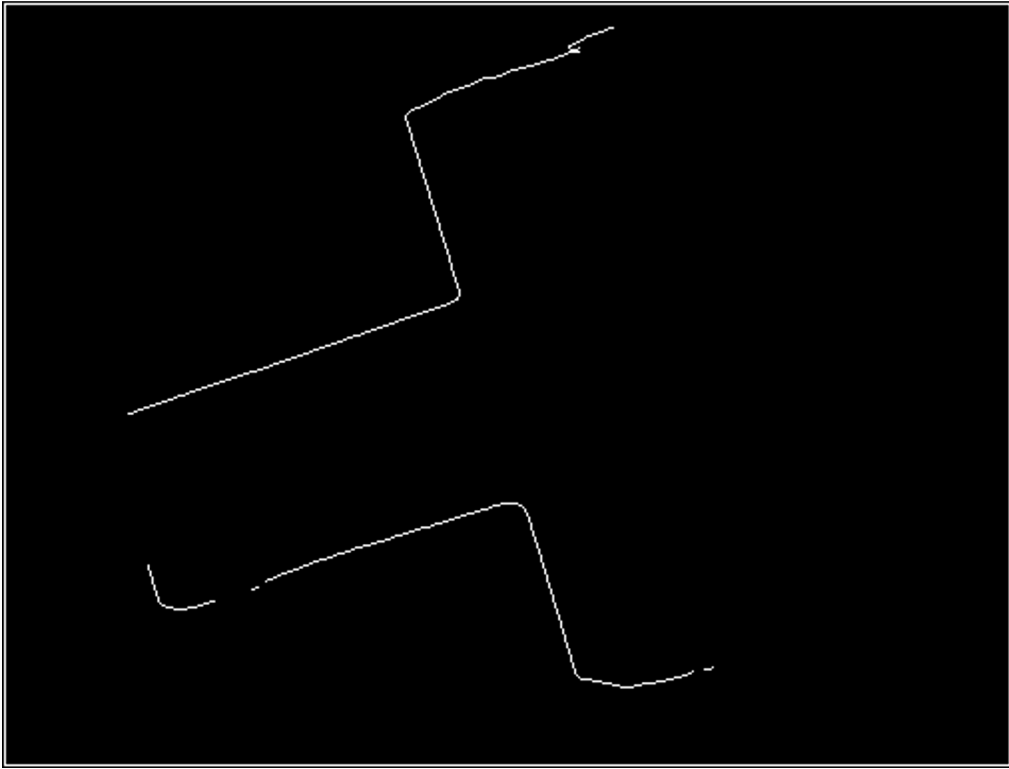


Edge Detection: Canny

- (1), (2), (3) ...
- (4) Simple thresholding (bottom left) versus Hysteresis thresholding (bottom right)

Weak Edge

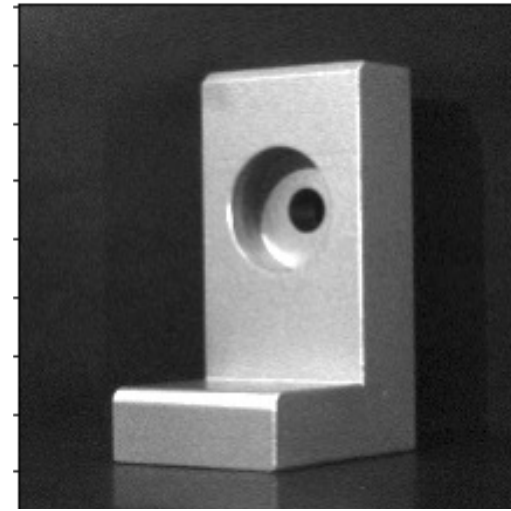
Very Weak Edge



Edge Detection: Canny

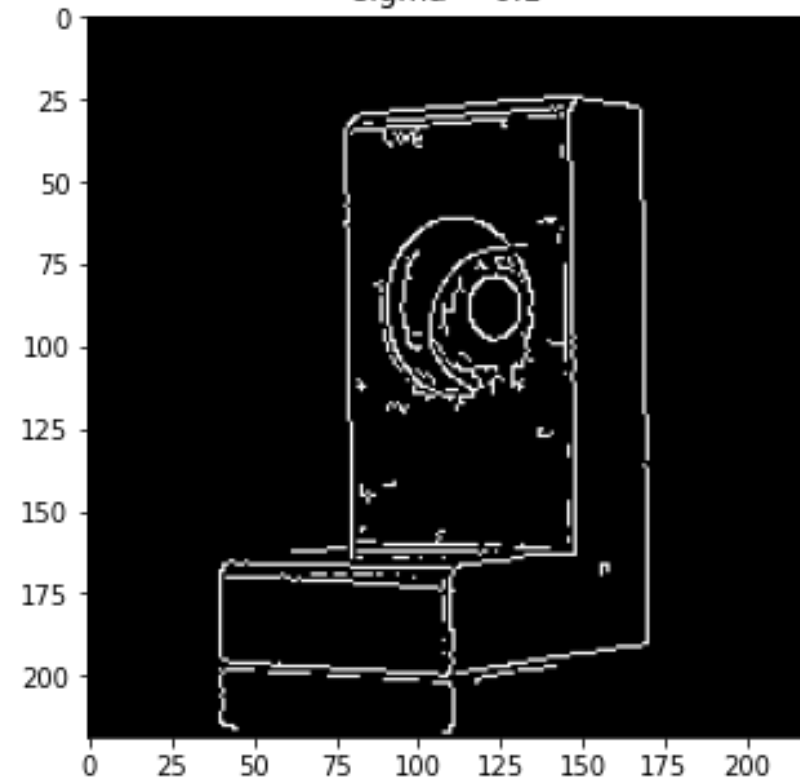
- Influence of **Gaussian standard deviation** (free-parameter 1 of 3)

Original image

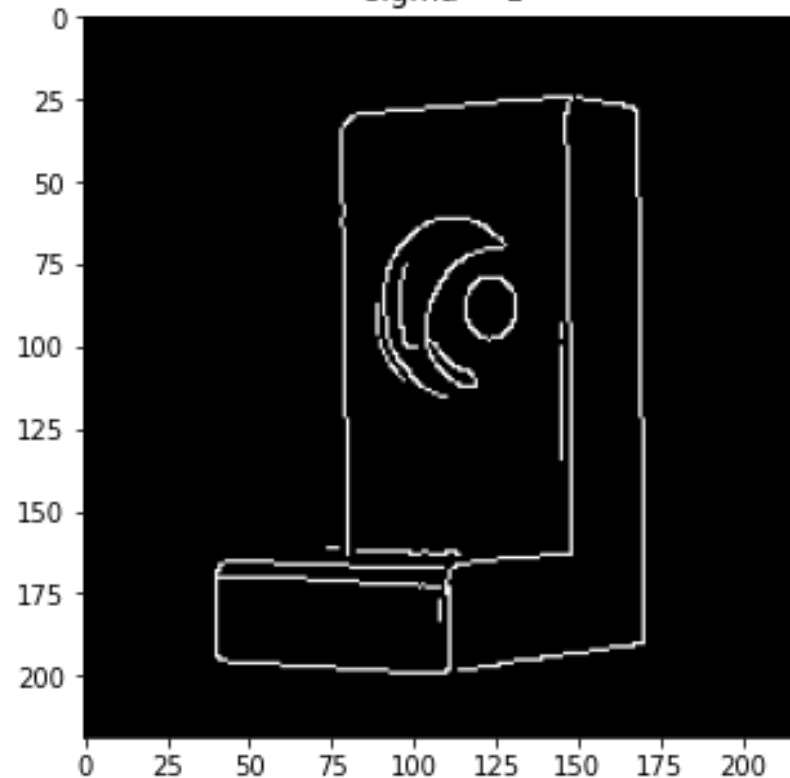


vincmazet.github.io/bip/labs/cor71.html

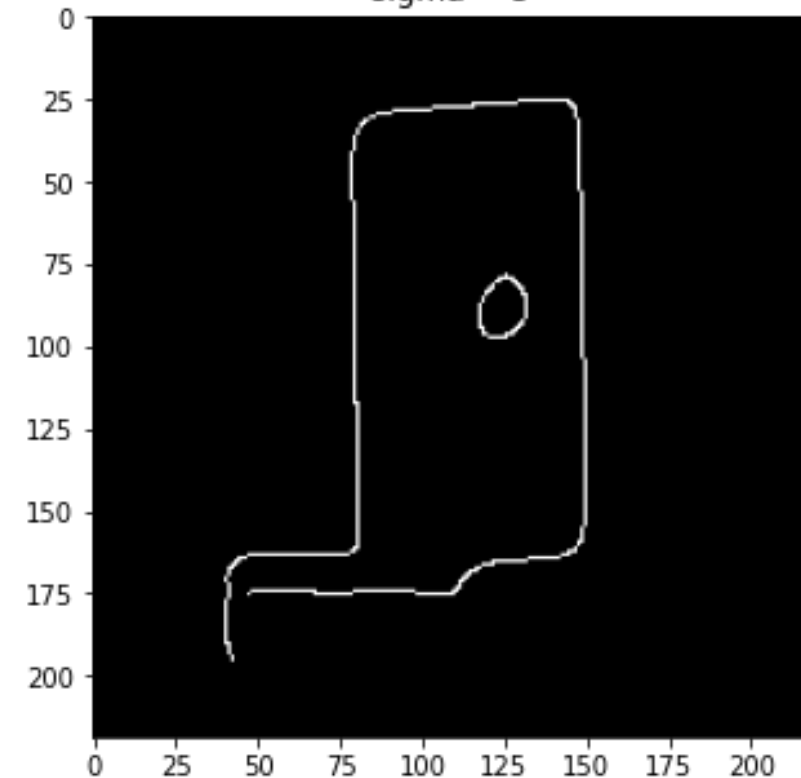
Canny detector
sigma = 0.1



Canny detector
sigma = 1



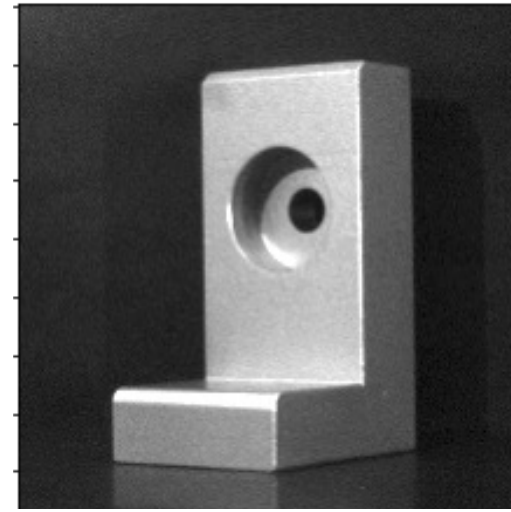
Canny detector
sigma = 5



Edge Detection: Canny

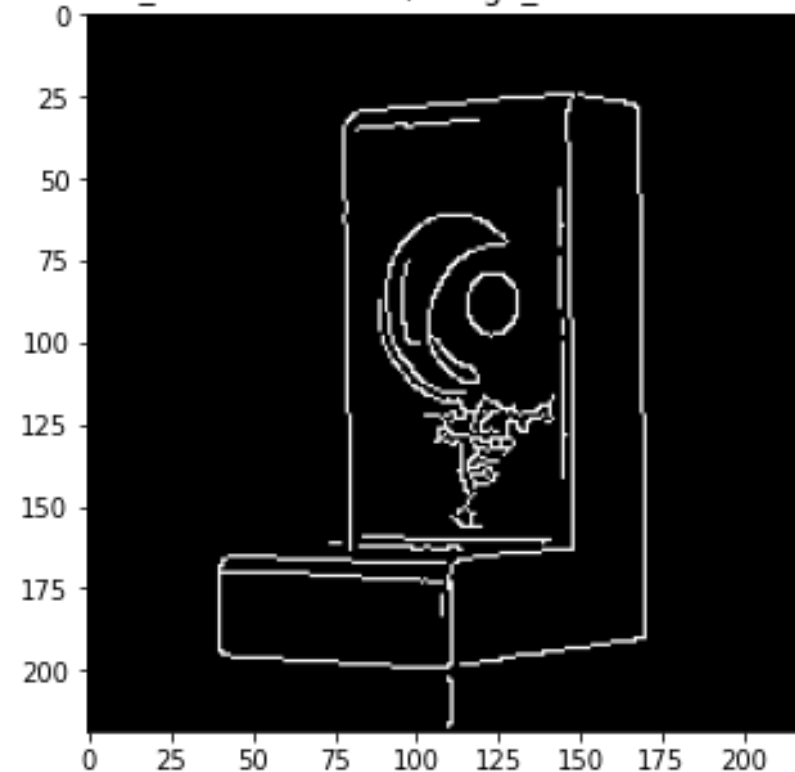
- Influence of **thresholdLow**
(free-parameter 2 of 3)

Original image



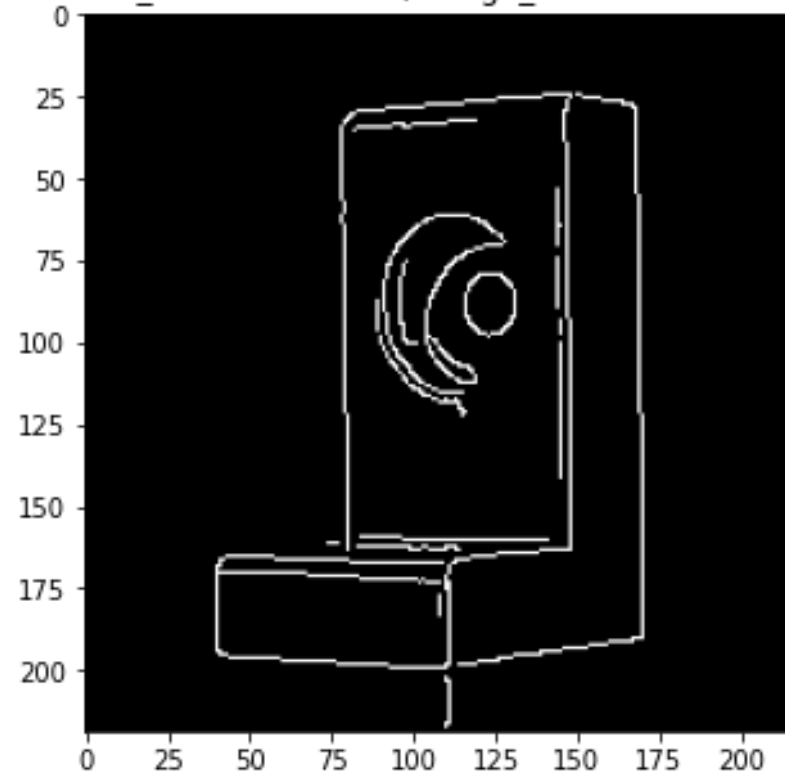
Canny detector

low_threshold = 1 %, high_threshold = 20 %



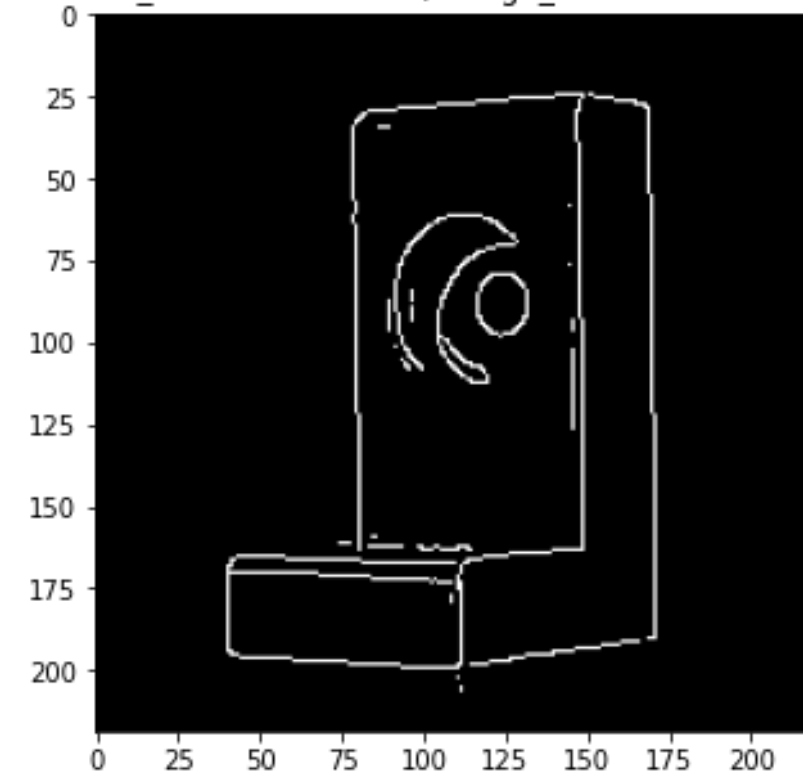
Canny detector

low_threshold = 5 %, high_threshold = 20 %



Canny detector

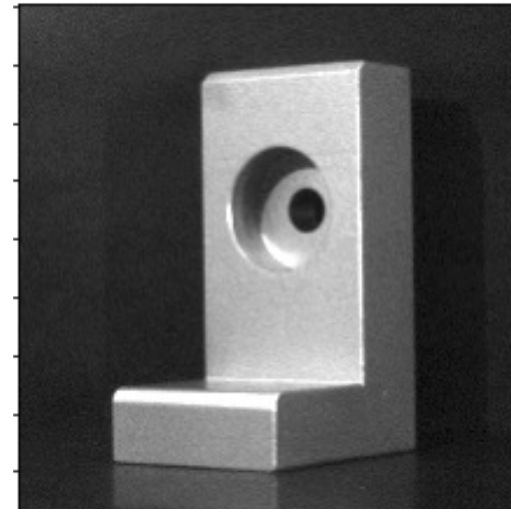
low_threshold = 20 %, high_threshold = 20 %



Edge Detection: Canny

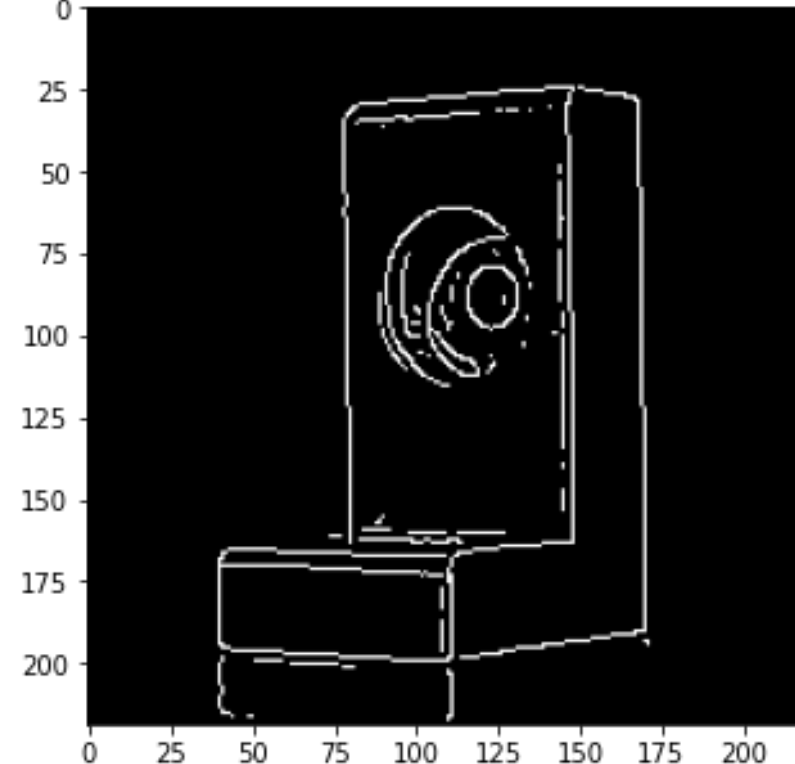
- Influence of **thresholdHigh**
(free-parameter 3 of 3)

Original image



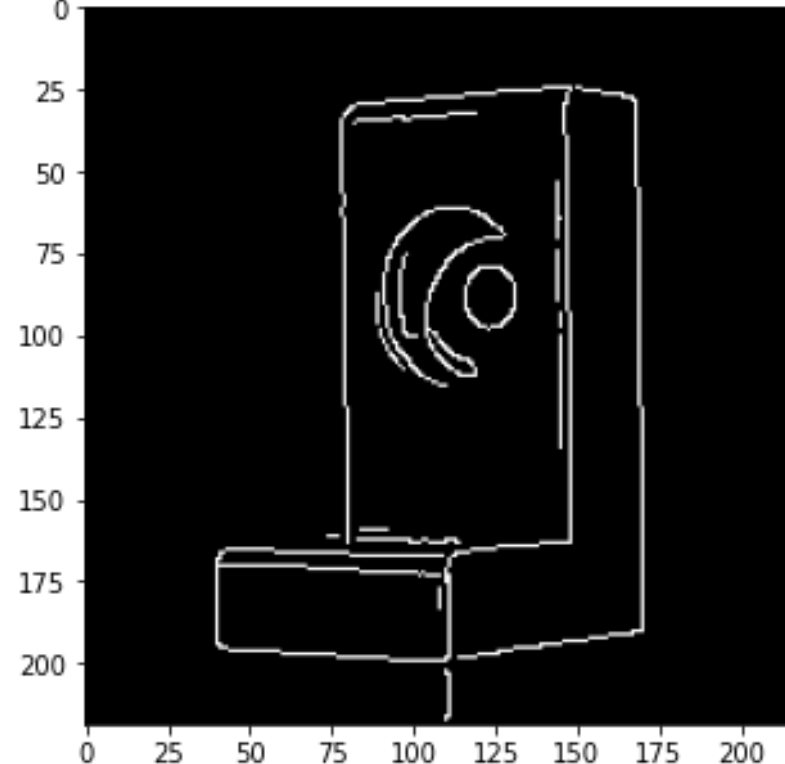
Canny detector

low_threshold = 10 %, high_threshold = 10 %



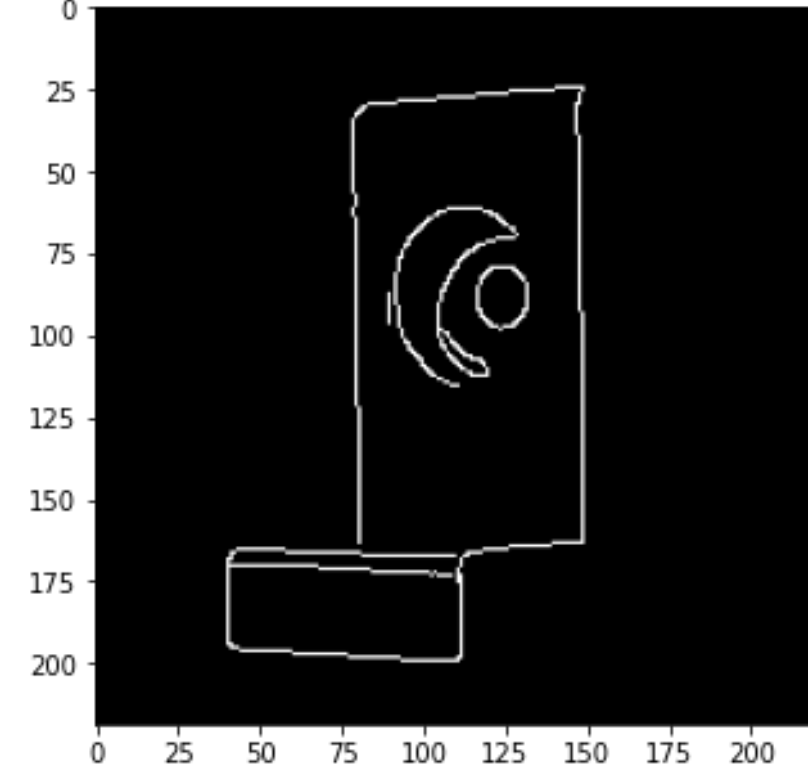
Canny detector

low_threshold = 10 %, high_threshold = 20 %



Canny detector

low_threshold = 10 %, high_threshold = 50 %



Edge Detection

- Comparing edge detectors

1) Sobel

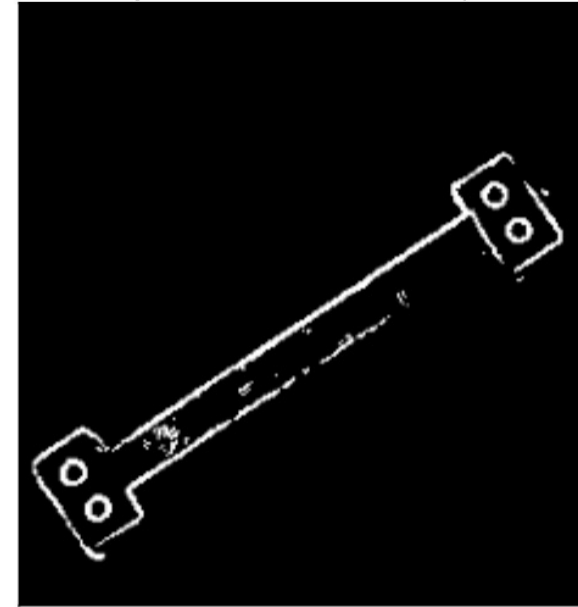
2) Marr-Hildreth

3) Canny

Original image



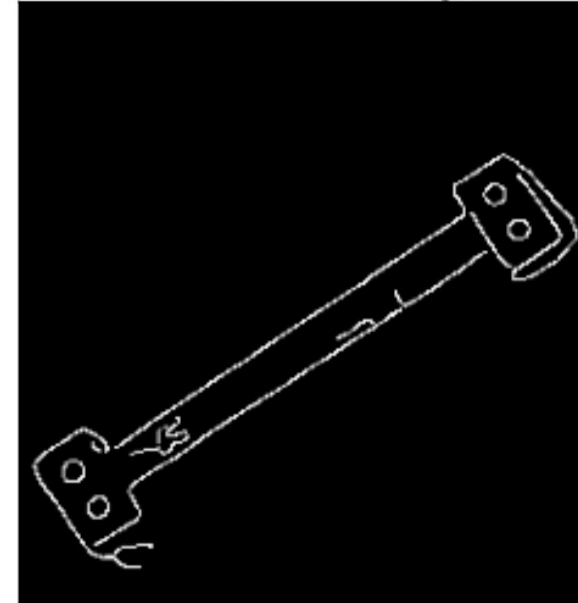
Sobel
(thresholded at 20)



Marr-Hildreth
($\sigma = 4$)

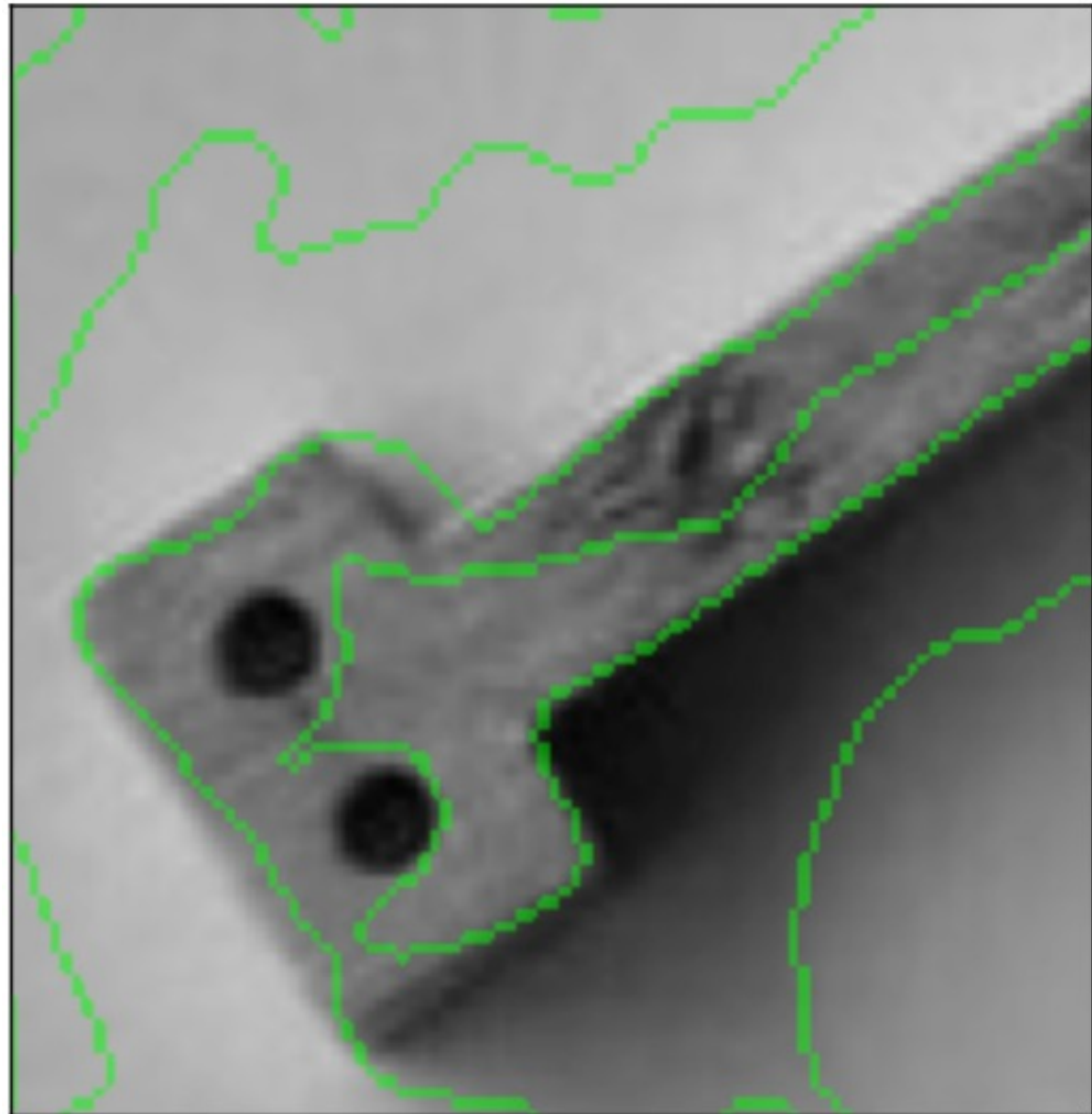


Canny
($\sigma = 1.5, T_{low} = 25, T_{high} = 50$)

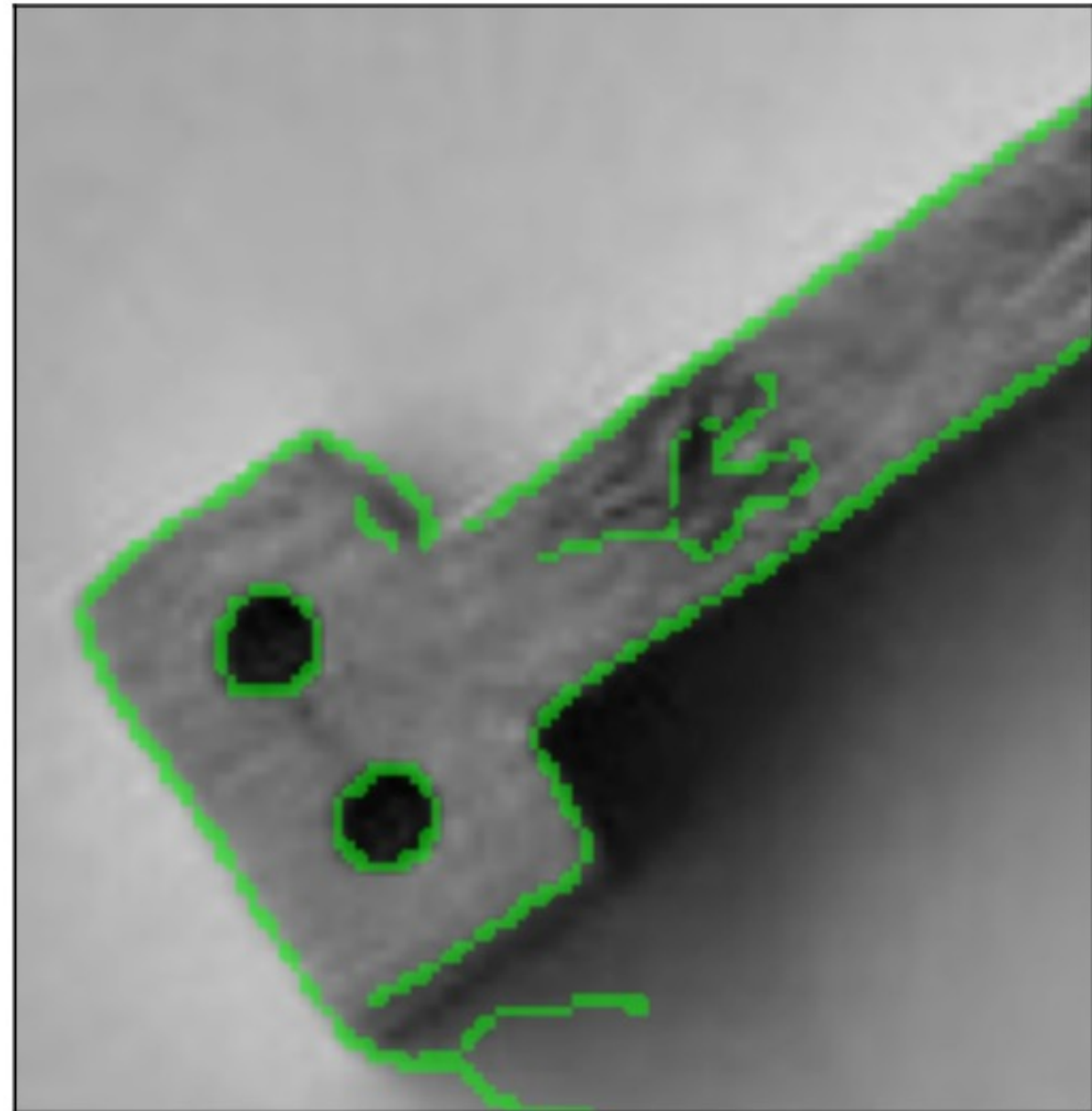


Edge Detection

Marr-Hildreth



Canny



Corner Detection: Moravec

- Motivation: track an object in a video, or in 2 images of same scene
- 1980



Obstacle Avoidance and Navigation in the Real World by a Seeing Robot Rover

Hans Moravec

Tech. Report, CMU-RI-TR-80-03, Robotics Institute, Carnegie Mellon University & doctoral dissertation, Stanford University, *September, 1980*

Corner Detection: Moravec

- Indicate pixel location “p” as a “corner” when there is low similarity (i.e., high dissimilarity) between:
 - (1) a patch around “p” and
 - (2) any/every other patch around neighbor location “q” of location “p”
- 3x3 neighborhood
- Threshold on similarity/dissimilarity
- Similarity/dissimilarity captured via
 - (1) sum of squared differences → dissimilarity OR
 - (2) (zero-normalized) cross correlation → similarity

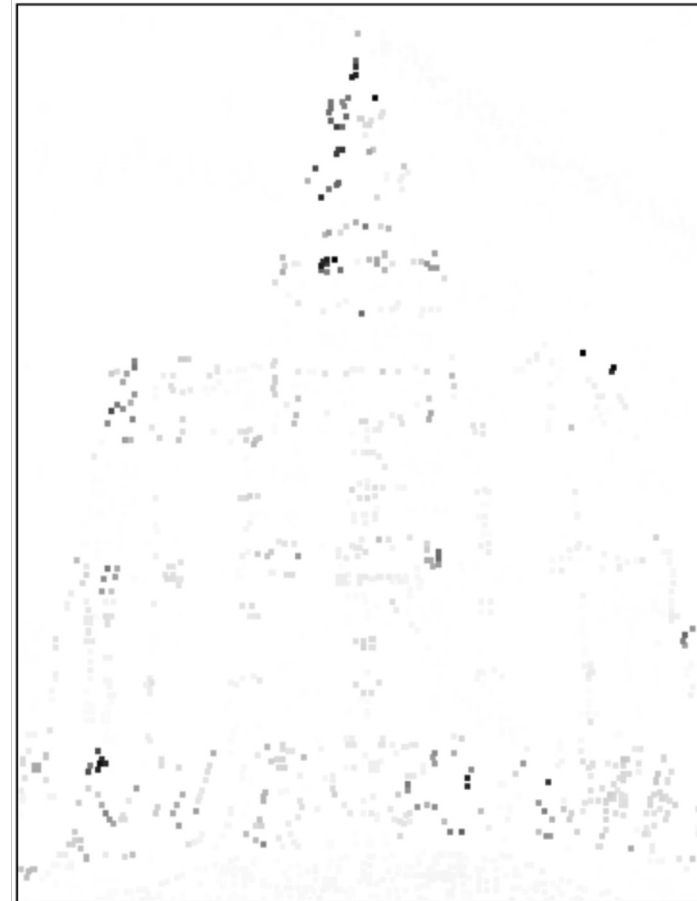
Corner Detection: Moravec

- Step 1/3 → At each “p”, for patch around “p”, compute similarity to patches around neighbors “q”
→ pick highest similarity, assign that value at location p
- White → high value
- Black → low value



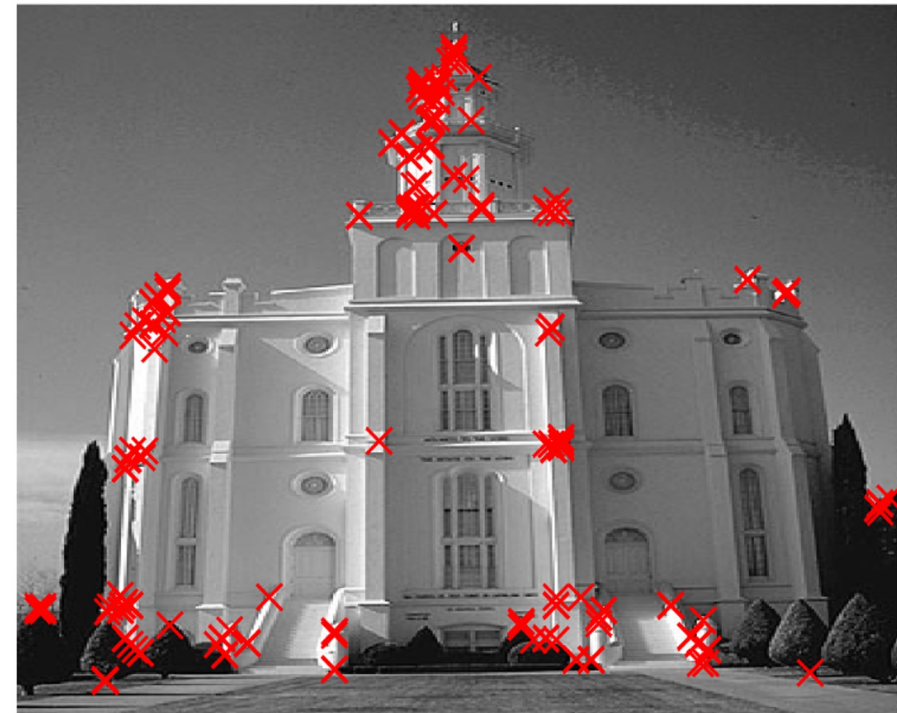
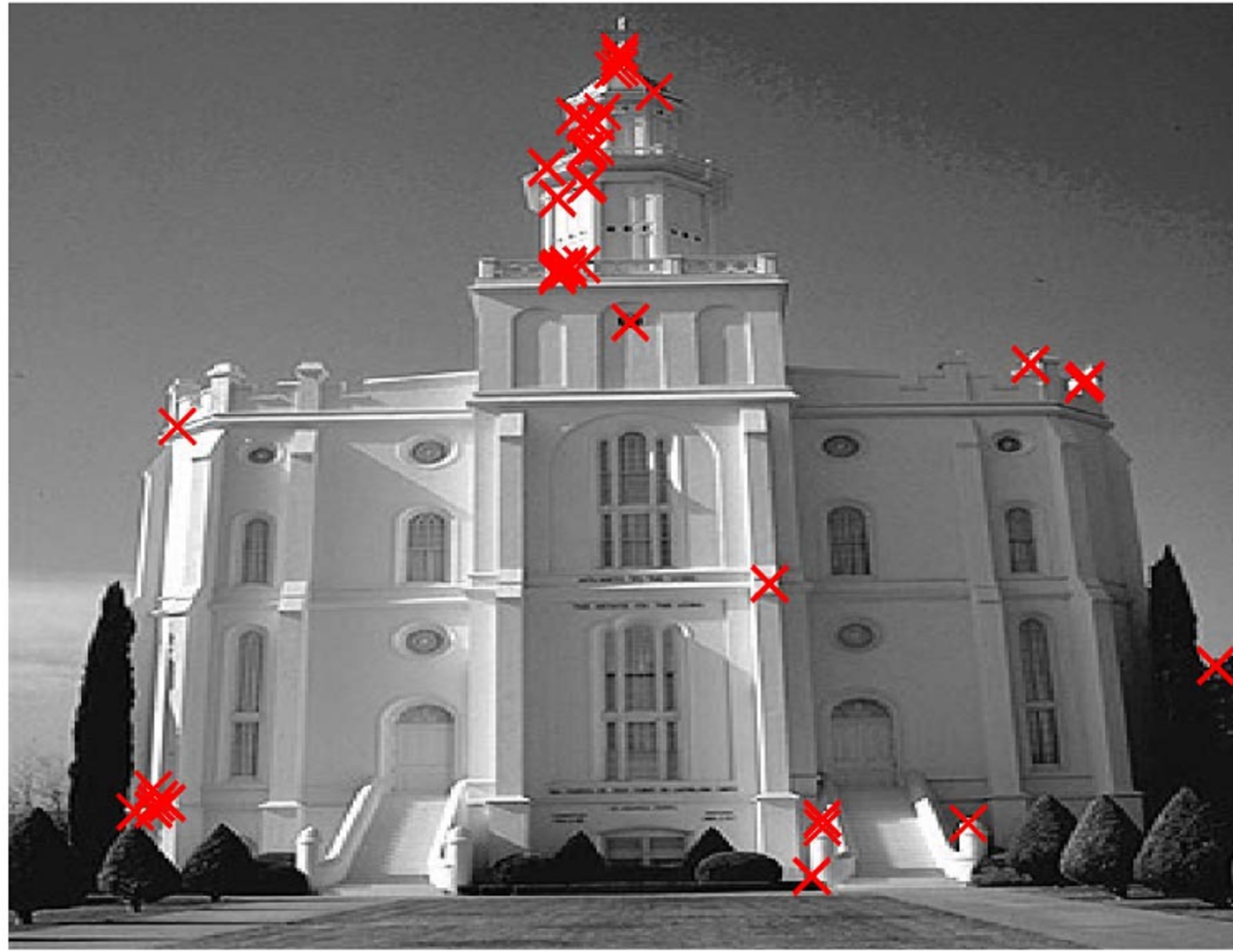
Corner Detection: Moravec

- Step 2/3 → Non-maximum suppression
- Ignore pixels having any 8-neighbor with lower values
- Because neighbor location is a better candidate for a corner
- Set such pixel's value to white



Corner Detection: Moravec

- Step 3/3 → Threshold



Corner Detection: Moravec

- Limitation
- Only 8 shifts of patches are considered at 45-degree angle increments
 - Left, right
 - Up, down
 - NW, NE, SW, SE
- Patches are rectangular (not isotropic)

Edge+Corner Detection: Harris-Stephens

A COMBINED CORNER AND EDGE DETECTOR

Chris Harris & Mike Stephens

Plessey Research Roke Manor, United Kingdom

© The Plessey Company plc. 1988

Edge+Corner Detection: Harris-Stephens

- Don't discretize shifts, analyze analytically to consider “all” (infinitely many) small (infinitesimally small) shifts
- Use Taylor-series expansion of image function
- Dissimilarity measured via sum of squared differences
- Dissimilarity between
 - (1) patch having pixel locations (u,v) and
 - (2) another patch over pixels **shifted** by (x,y)

$$S(x, y) = \sum_u \sum_v w(u, v) [I(u + x, v + y) - I(u, v)]^2$$

- where $w(u,v) :=$ weight for location (u,v) within patch, e.g., Gaussian, can be used to make patch isotropic or disc-shaped

Edge+Corner Detection: Harris-Stephens

$$S(x, y) = \sum_u \sum_v w(u, v) [I(u + x, v + y) - I(u, v)]^2$$

- Taylor-series expansion of image function around location (u,v):

$$I(u + x, v + y) \approx I(u, v) + I_x(u, v)x + I_y(u, v)y$$

- Thus, approximated patch dissimilarity is:

$$S(x, y) \approx \sum_u \sum_v w(u, v) [I_x(u, v)x + I_y(u, v)y]^2$$

- Interpret this as quadratic form over shift variables x,y

- Patch dissimilarity at any shift (x,y) is: $S(x, y) \approx \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix}$

Edge+Corner Detection: Harris-Stephen

$$S(x, y) \approx \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix}$$

- A = local **structure tensor**/matrix in patch with locations (u,v)

$$A = \sum_u \sum_v w(u, v) \begin{bmatrix} I_x(u, v)^2 & I_x(u, v)I_y(u, v) \\ I_x(u, v)I_y(u, v) & I_y(u, v)^2 \end{bmatrix} = \begin{bmatrix} \langle I_x^2 \rangle & \langle I_x I_y \rangle \\ \langle I_x I_y \rangle & \langle I_y^2 \rangle \end{bmatrix}$$

- Tensor A gives averaged local differential structure
- Tensor A is symmetric, positive (semi-)definite

- Because each summand matrix at a particular location (u,v) $\rightarrow \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}_{(u,v)} = \begin{bmatrix} I_x \\ I_y \end{bmatrix}_{(u,v)} \cdot \begin{bmatrix} I_x & I_y \end{bmatrix}_{(u,v)}$

- Parameters:

Gaussian for neighborhood specification $w(u,v)$

Gaussian smoothing for computing I_x, I_y

Edge+Corner Detection: Harris-Stephen

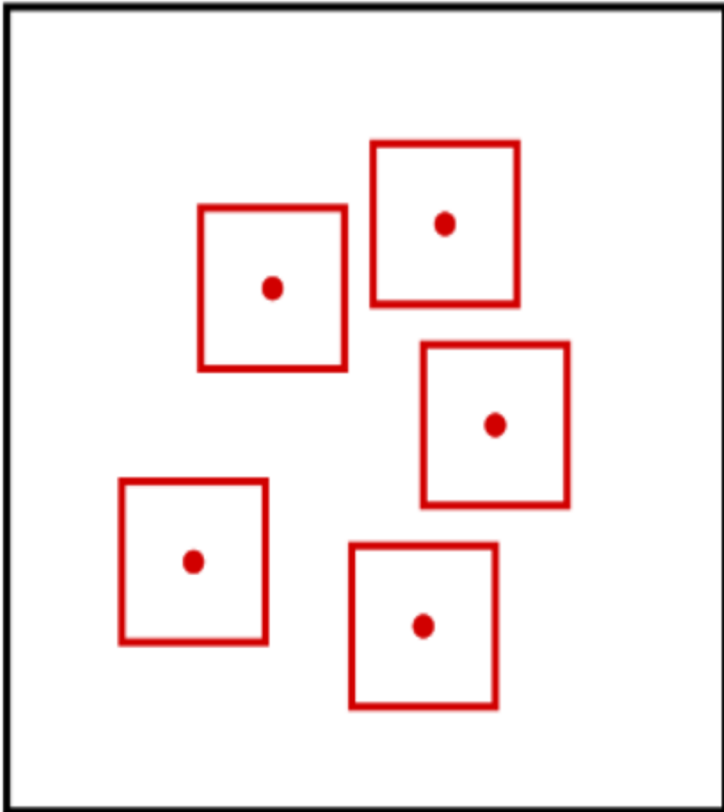
$$S(x, y) \approx \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix}$$

- A has an eigen decomposition: $A = Q D Q^{-1} = V$
- Eigenvalues $\rightarrow$ non-negative
- Eigenvectors $\rightarrow$ orthogonal
- For a patch's center pixel to be at an object's corner, dissimilarity $S(x,y)$ should be large for all shifts (x,y)
- Thus, "A" must have 2 large (positive) eigenvalues
 - Why ?
- If A has an eigenvalue = 0, then for shift (x,y) along corresponding eigenvector "v",
 $v' A v = 0$

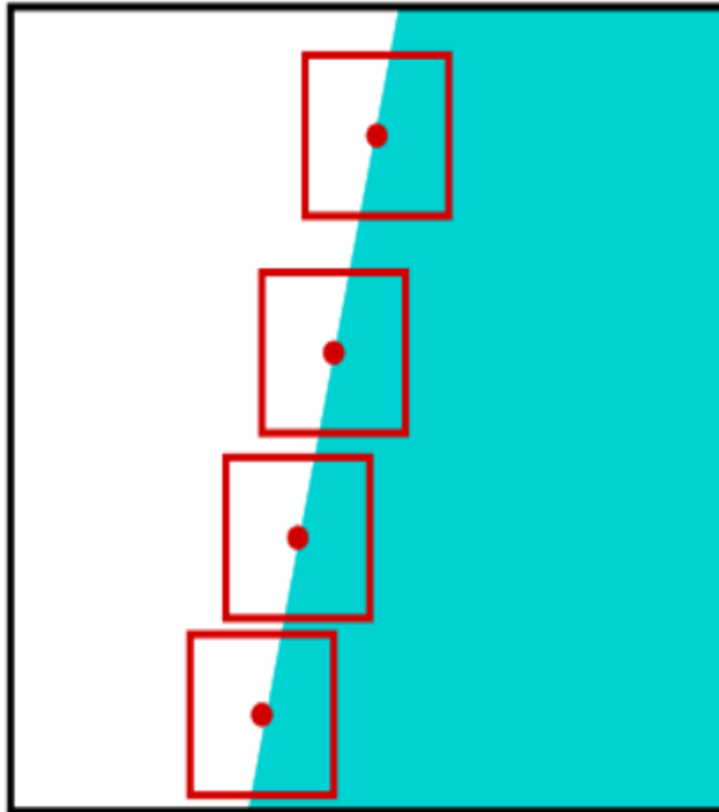
Edge+Corner Detection: Harris-Stephens

- 1) Constant region: both eigenvalues = 0
- 2) Edge: one eigenvalue large, other $\rightarrow 0$
- 3) Corner: both eigenvalues large

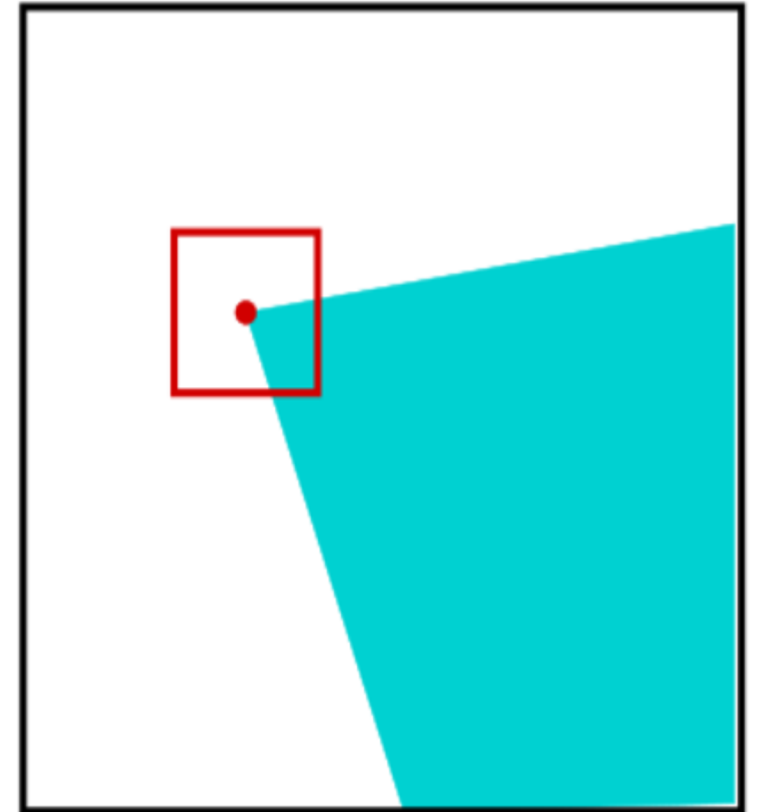
flat region



edge



corner



Edge+Corner Detection: Harris-Stephens

- Corner and edge measure designed to avoid eigen decomposition
- **Corner-ness C** := $\text{Determinant}(A) - k (\text{Trace}(A))^2$
 - $\text{Determinant}(A)$ = product of eigenvalues
 - $\text{Trace}(A)$ = sum of eigenvalues = sum of diagonal elements
 - k := empirically-tuned constant **such that following holds:**
 - 1) If both eigenvalues are close to 0, then $C \rightarrow 0$ (flat case)
 - 2) If one eigenvalue is close to 0, and other is large, then $C < 0$ (edge case)
 - 3) If both eigenvalues large, then $C > 0$ (corner case)
 - Analysis of limits on k (see next slide)

[Case 1] $k \leq 0 \rightarrow C \geq 0$

[Case 2] $k \geq 0.25 \rightarrow C \leq 0$
 - Rotation invariant: because eigenvalues don't change when image is rotated, i.e., when gradient vectors $\langle I_x, I_y \rangle$ get rotated
 - **Tune k within (0, 0.25)**

Edge+Corner Detection: Harris-Stephens

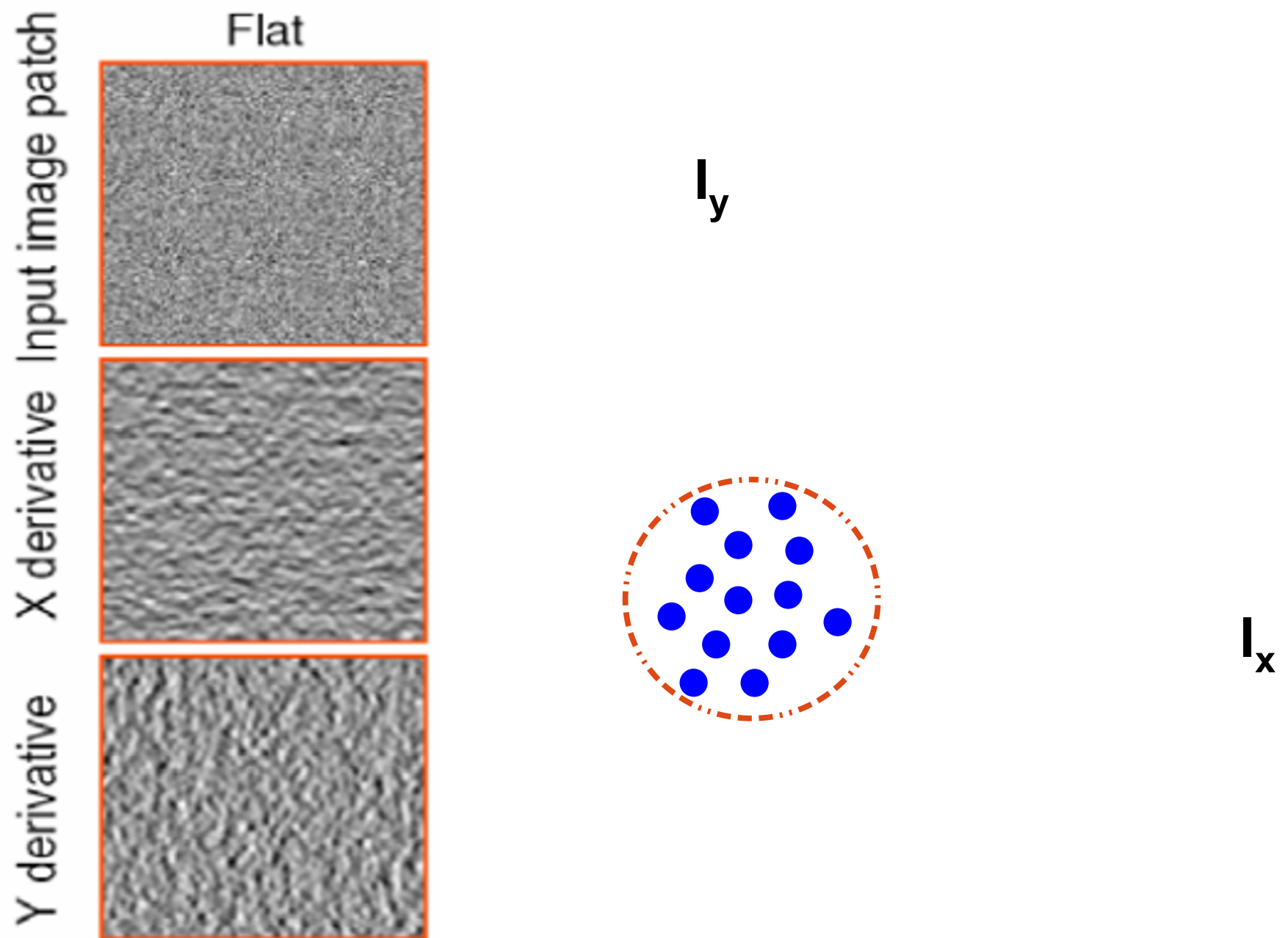
- For 2D images: $C = ab - k(a+b)^2 = ab + k(-a^2 - b^2 - 2ab)$
 - We know that a, b are non-negative eigenvalues
- When $k := 0.25 + x$, where x is small ($|x| \ll 1$):
 - Then, $C = 0.25(-a^2 - b^2 + 2ab) + x(-a^2 - b^2 - 2ab)$
 - Then, $C = -0.25(a-b)^2 - x(a+b)^2$
 - Then, $x \geq 0 \rightarrow C \leq 0$ (for any non-negative a, b) $\rightarrow$ non-informative
 - Then, $x < 0 \rightarrow$ there exist some a, b for which $C > 0$, e.g., $a = b \gg 0$ (corner)
- When $k := 0 + y$, where y is small ($|y| \ll 1$):
 - Then, $C = ab - y(a+b)^2$
 - Then, $y \leq 0 \rightarrow C \geq 0$ (for any non-negative a, b) $\rightarrow$ non-informative
 - Then, $y > 0 \rightarrow$ there exist some a, b for which $C < 0$, e.g., $a \gg b = 0$ (edge case)
- **Tune k within $(0, 0.25)$ to separate corners from edges**

Edge+Corner Detection: Harris-Stephens

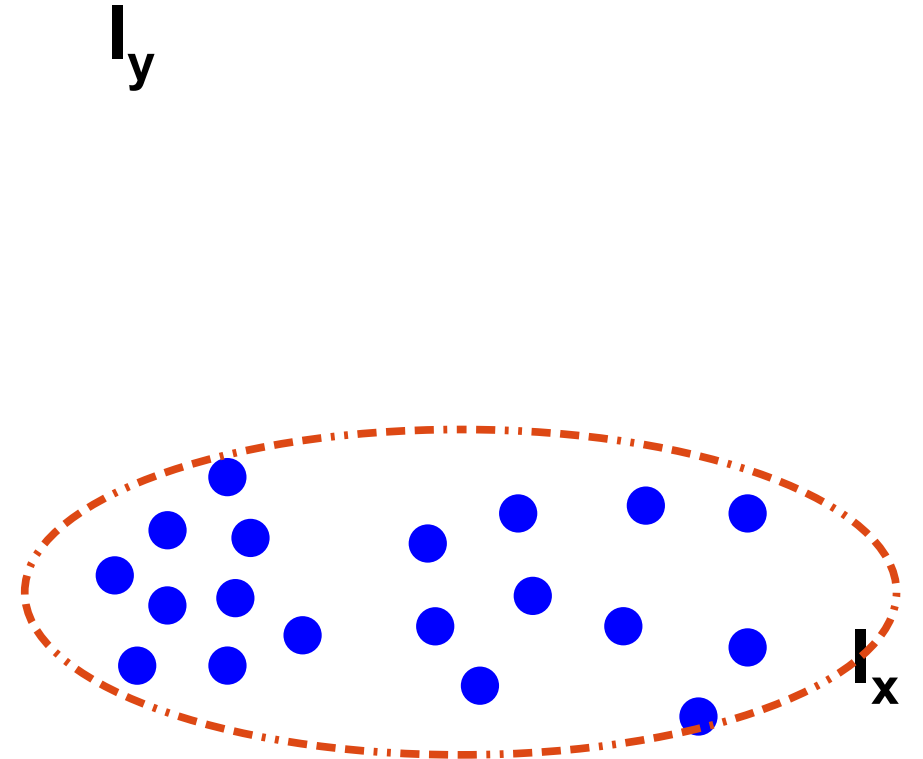
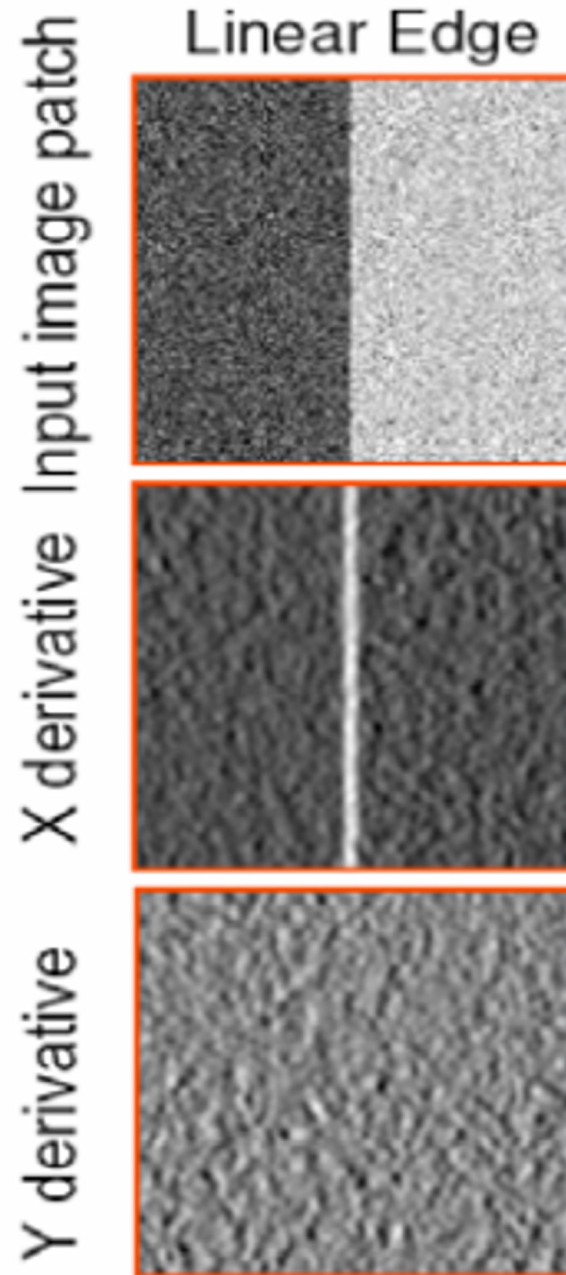
- Another interpretation of “A”
- Ignoring weights $w(u,v)$,
“A” = 2nd-moment matrix of gradient vectors over patch
- Not covariance matrix, because no mean subtraction

$$A = \sum_u \sum_v w(u, v) \begin{bmatrix} I_x(u, v)^2 & I_x(u, v)I_y(u, v) \\ I_x(u, v)I_y(u, v) & I_y(u, v)^2 \end{bmatrix} = \begin{bmatrix} \langle I_x^2 \rangle & \langle I_x I_y \rangle \\ \langle I_x I_y \rangle & \langle I_y^2 \rangle \end{bmatrix}$$

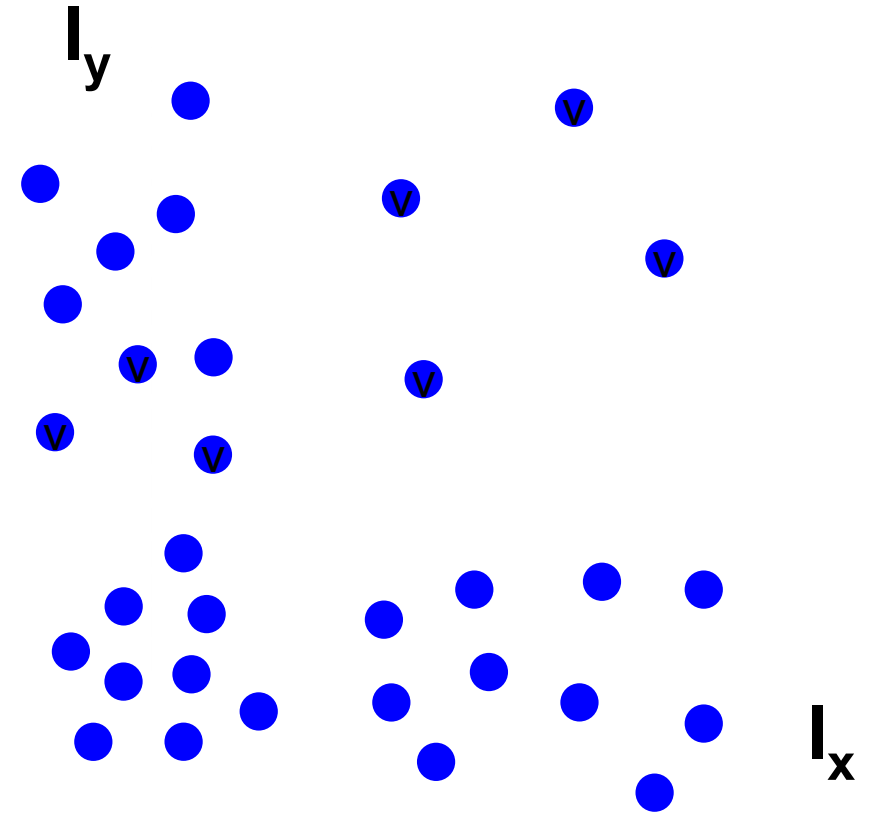
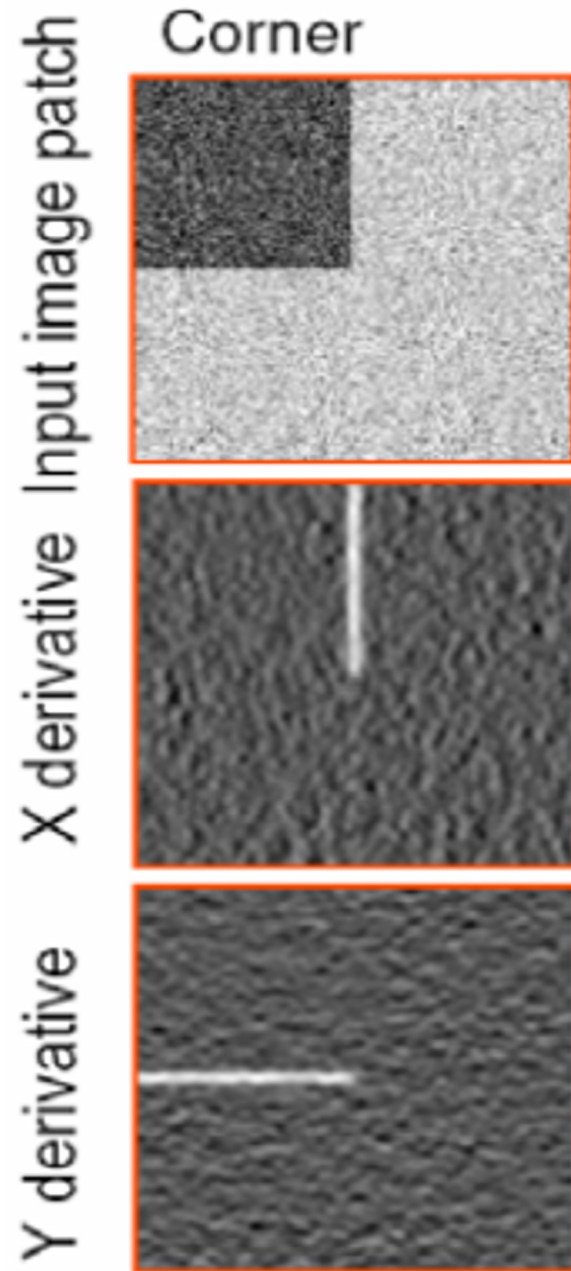
Edge+Corner Detection: Harris-Stephens



Edge+Corner Detection: Harris-Stephens



Edge+Corner Detection: Harris-Stephens



Edge+Corner Detection: Harris-Stephens

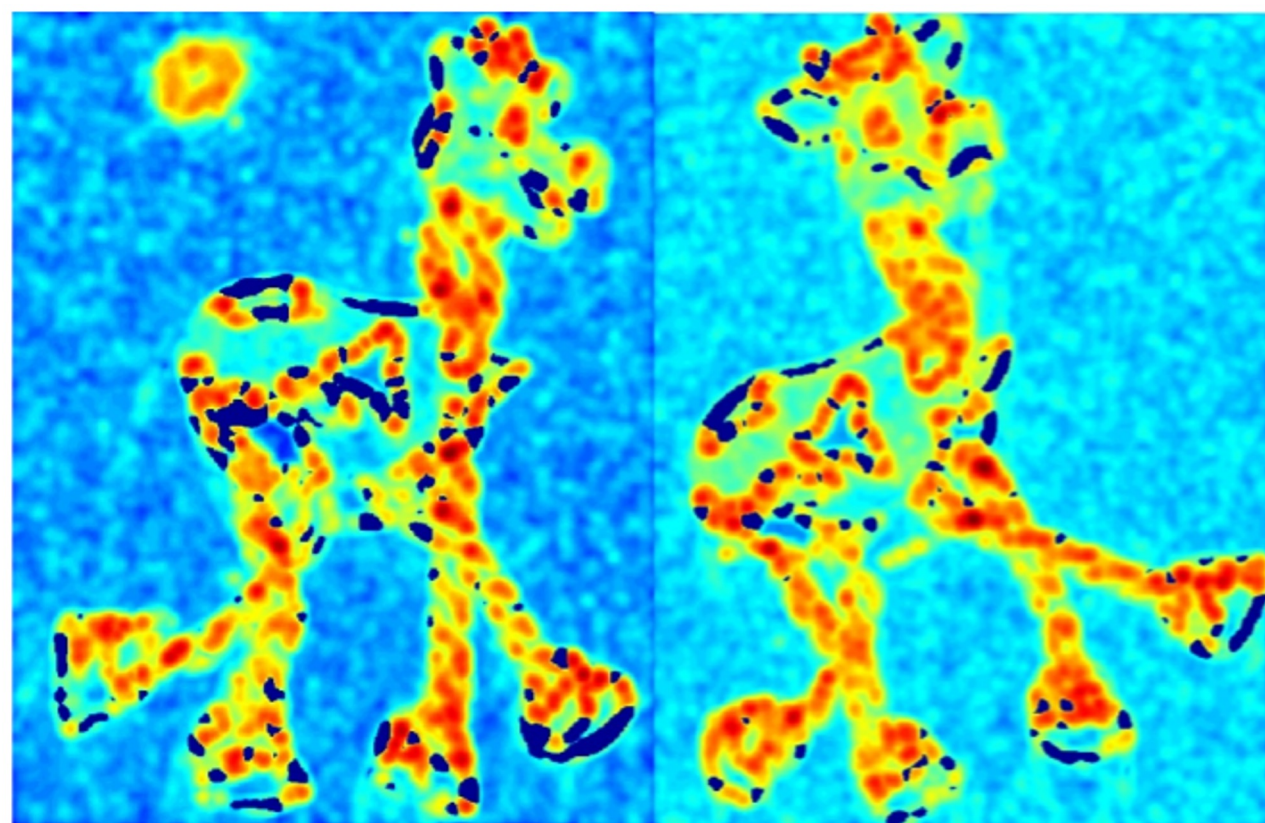
- Challenging examples
 - 1) Compute corner-ness measure
 - 2) Non-max suppression (in neighborhoods)



Edge+Corner Detection: Harris

1) Corner-ness measure

- Red → large positive
- Blue → large negative



Edge+Corner Detection · Harris-Stenhens

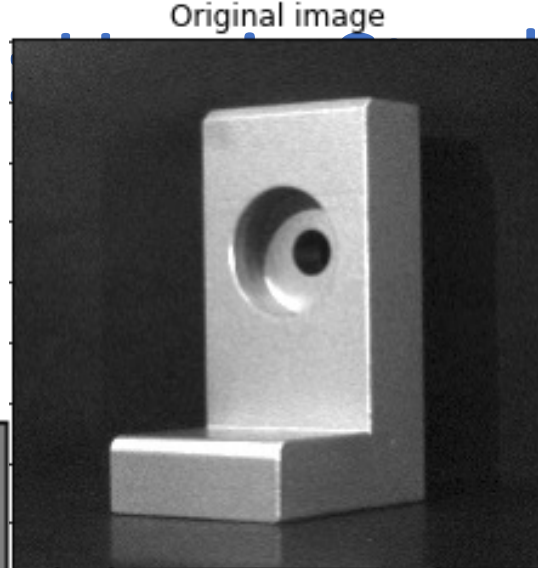
2) Non-max suppress



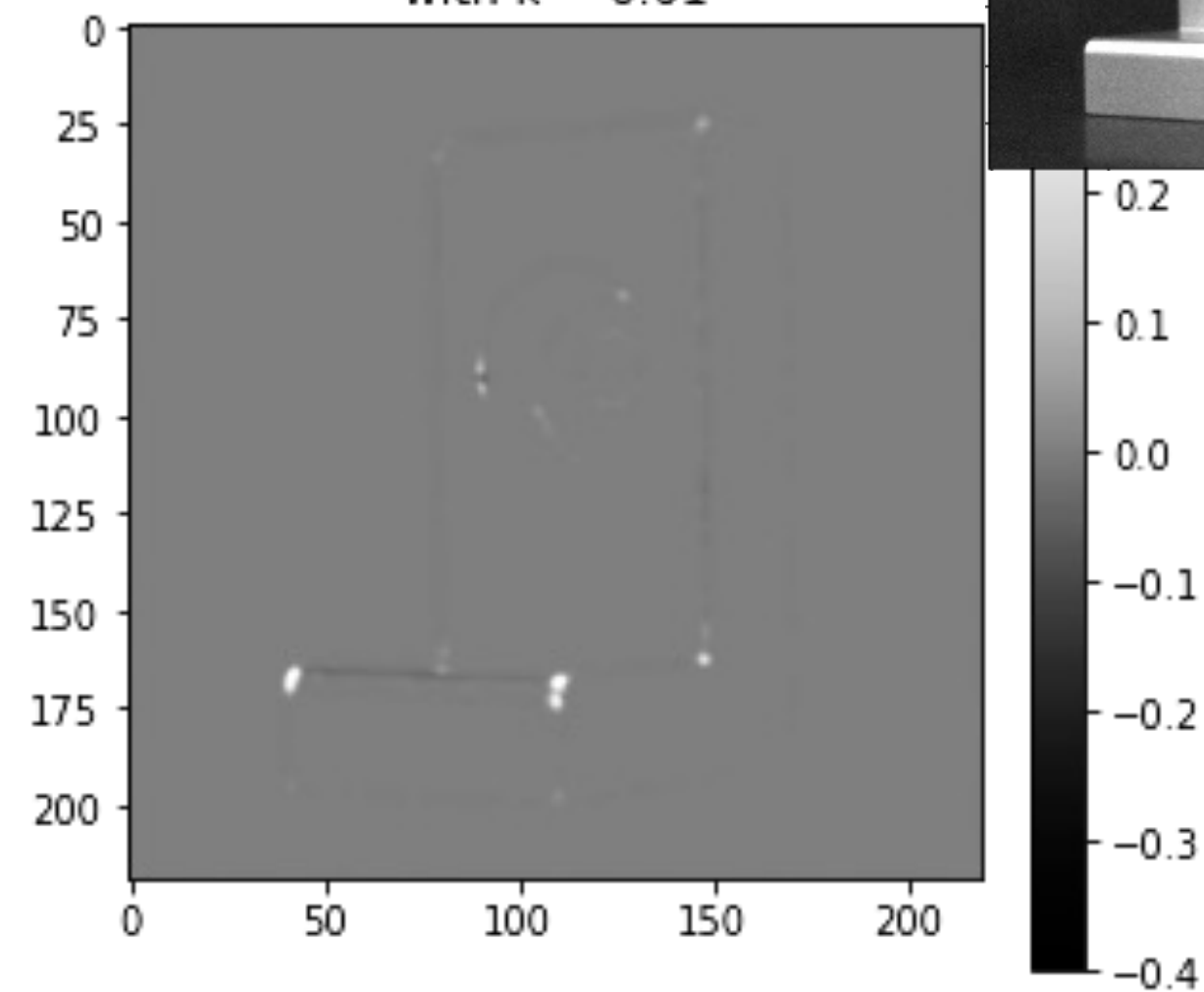
Edge+Corner Detection: Harris-Stephe



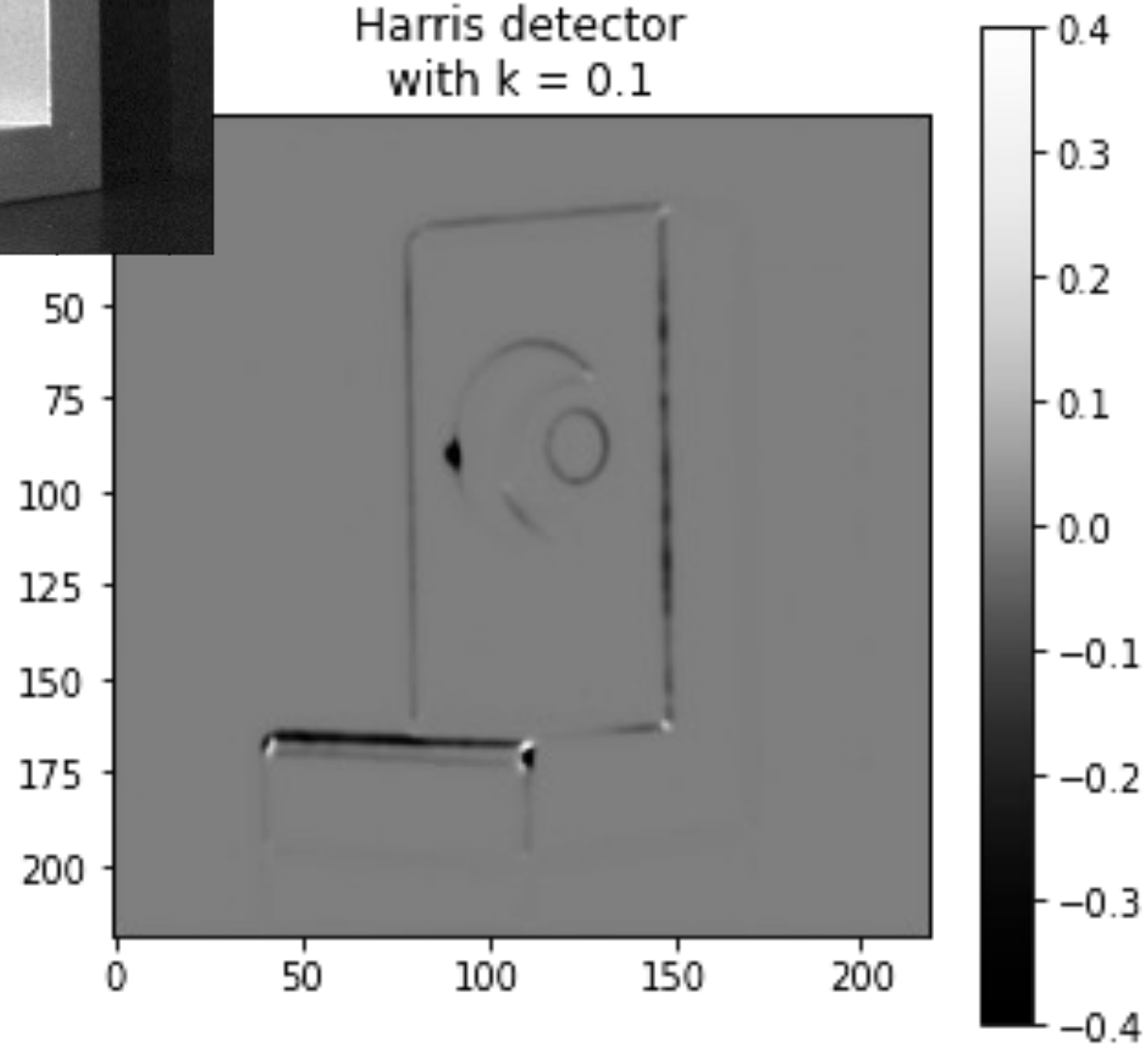
Edge+Corner Detection with Harris



Harris detector
with $k = 0.01$

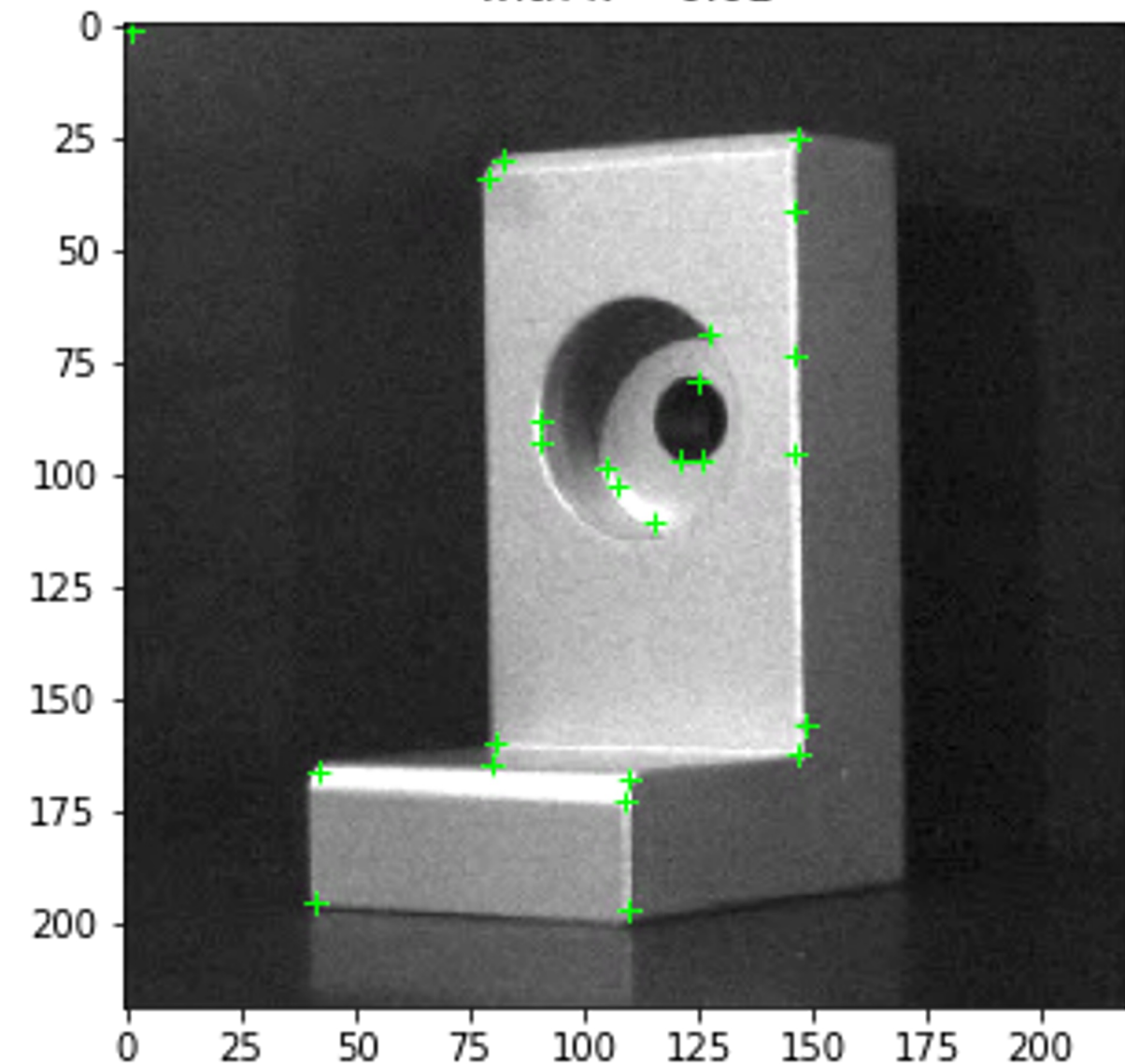


Harris detector
with $k = 0.1$

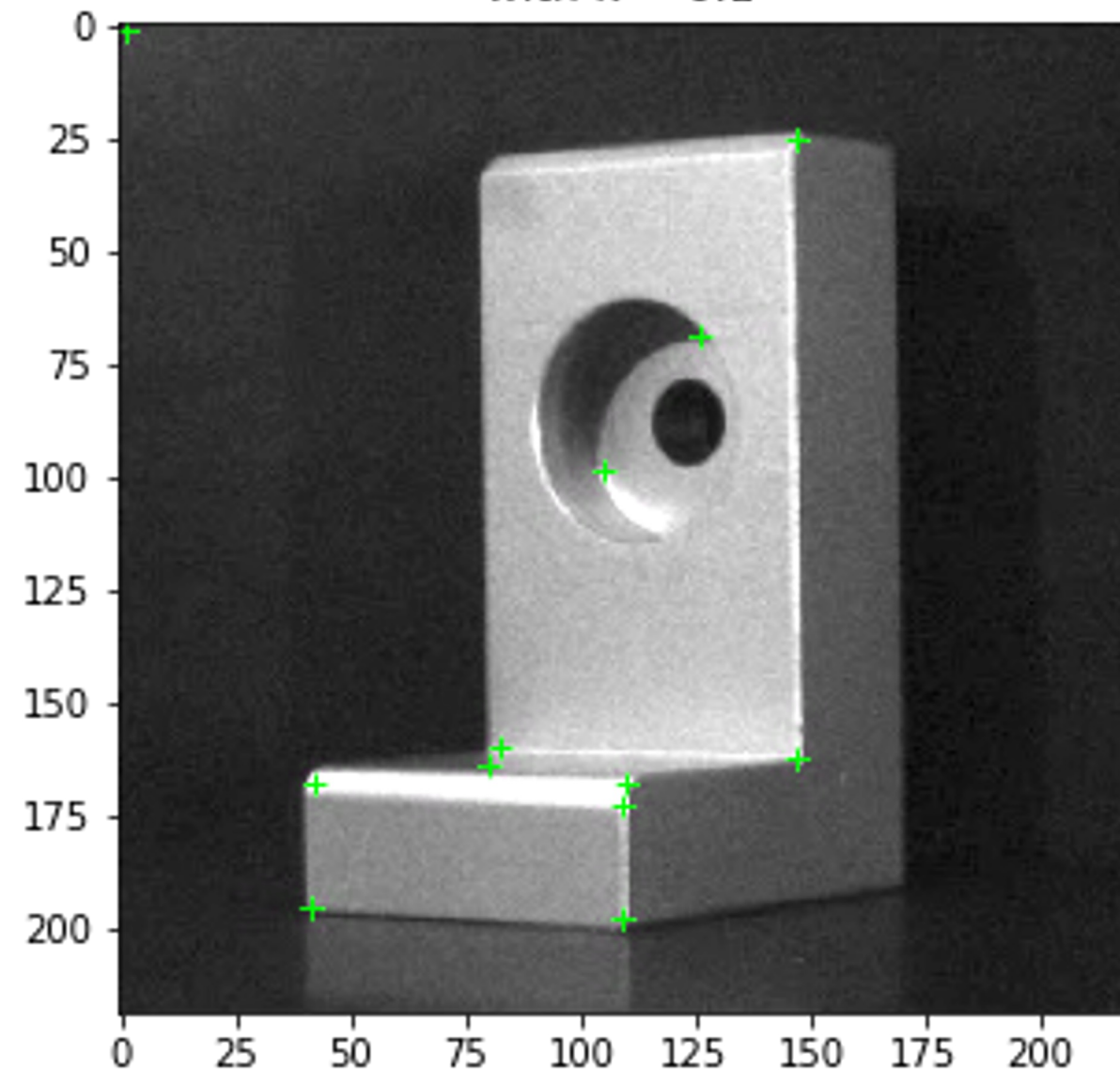


Edge+Corner Detection: Harris-Stephens

Result given by the function `corner_peaks`
with $k = 0.01$



Result given by the function `corner_peaks`
with $k = 0.1$



Corner Detection: Shi-Tomasi 1994

- Need features to track across images of same scene (e.g., in video)
- Can use points with 2 large eigenvalues of structure tensor, indicating corners or other complex features
- **Feature/Corner Score := min of 2 eigenvalues**
 - Min can be computed in closed form for 2x2 matrix
- Use non-maximum suppression
- Tracking criterion: Feature/Corner Score > threshold

IEEE Conference on Computer
Vision and Pattern Recognition
(CVPR94) Seattle, June 1994

Good Features to Track

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ieeexplore.ieee.org/document/323794

Carlo Tomasi

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Corner Detection: Brown-Szeliski-Winder 2005

- Corner scores at multiple scales
- At scale l , “Harris matrix” = $\mathbf{H}_l(x, y) = \nabla_{\sigma_d} P_l(x, y) \nabla_{\sigma_d} P_l(x, y)^T * g_{\sigma_i}(x, y)$
- At scale l , **interest-point/corner score** = $\frac{\det \mathbf{H}_l(x, y)}{\text{tr} \mathbf{H}_l(x, y)} = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
- Use non-maximum suppression
- They also compute **orientation** $\leftarrow$ smoothed local gradient

Multi-Image Matching using Multi-Scale Oriented Patches

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*Vision Technology Group
Microsoft Research*

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ieeexplore.ieee.org/document/1467310

Corner Detection: Brown-Szeliski-Winder 2005

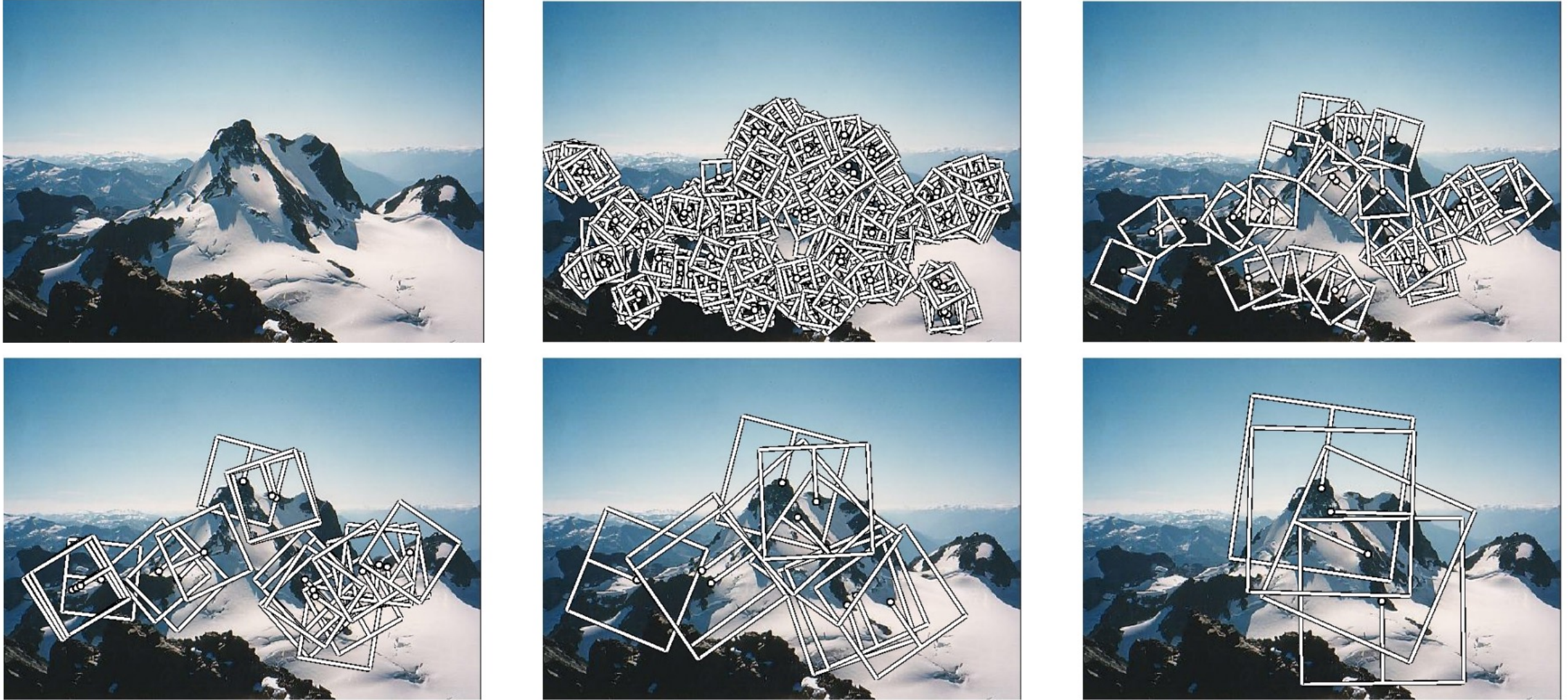


Figure 1. Multi-scale Oriented Patches (MOPS) extracted at five pyramid levels from one of the Matier images. The boxes show the feature orientation and the region from which the descriptor vector is sampled.